

Fragile Fortuity: The Gravity of BPS Overlaps in Supersymmetric SYK

Abstract

The fortuitous BPS states of the $\mathcal{N}=2$ SYK model are strongly chaotic, and their random-matrix description has recently been developed on the *energy-spectrum* side. We develop the complementary *overlap* side. The uplift decoder $O_T = P_B \Pi_T P_B$, which measures the metric fragility of BPS states across $N \rightarrow N+1$, is a Gram matrix of gravitational overlaps whose disorder-averaged spectrum is the two-projector free-probability (Wachter/MANOVA) law—a two-hard-edge (Jacobi) counterpart of the one-hard-edge (Bessel/Marchenko–Pastur) energy models. We realize it as a Penington–Shenker–Stanford–Yang sum over end-of-the-world branes in the topological ($E = 0$) sector of $\mathcal{N}=2$ super-JT gravity: the genus expansion of the overlap moments *is* the gravitational sum over bulk topologies, the disk gives Wachter, and the whole handle tower resums into a closed-form genus-one resolvent $W_1(x) = -\gamma^2 x(x-1)/(a\sigma^5)$; the amplitude dictionary follows from the topological term $e^{S_0\chi}$ of the action, so the gravity reproduces the handles to all orders. Two features escape this universal description. (i) A protected-atom resonance at $\lambda = 0, 1$ —exactly transmitted/lost BPS directions in excess of the ensemble—which is a finite- N effect ($\sim 5\%$ at $N = 12$ – 13 , vanishing by $N = 14$), distinct from the robust $\mathbb{Z}_q$ -Witten index $3^{(N-1)/2}$. (ii) A *macroscopic soft tail*: the exact decoder places an $O(10\%)$, non-vanishing fraction of BPS states below the Wachter hard edge, where the ensemble (hence the wormhole sum) predicts none—a genuine breakdown of random-matrix universality, distinct from the dilute sub-gap tunneling of the energy models. We trace this tail to its origin: it is the smaller theory’s near-BPS spectrum imaged into overlap variables by the uplift. The map is an exact identity $\lambda = E/(E + \sigma_a^2)$ derived from the BPS constraint, which for near-BPS energies is linear, $\lambda \simeq cE$ (correlation 0.95, c predicted from the couplings) —the overlap shadow of the near-extremal throat, tying the tail directly to the energy-side model. Finally, the chaotic (fortuitous) states carry Gaussian-unitary statistics while the monotone tower is deterministic, so the decoder separates black-hole-microstate from graviton-gas content. The dual is one of *gravitational overlaps*, not of spacetime geometry.

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1 The decoder side

1.1 Model and the uplift decoder

We work with the generic one-flavor $\mathcal{N}=2$ SYK model in its wedge realization: the Hilbert space is the exterior algebra $\mathcal{H} = \Lambda^\bullet(\mathbb{C}^N)$, graded by fermion number p with $\dim \mathcal{H}^p = \binom{N}{p}$, and the supercharge is wedging with a generic q -form ($q = 3$),

$$Q = C \wedge (\cdot), \quad C = \sum_{i < j < k} C_{ijk} e_i \wedge e_j \wedge e_k, \quad Q^2 = 0, \quad H = \{Q, Q^\dagger\}. \quad (1)$$

The BPS states are the harmonic representatives $\mathcal{B}^p = \ker Q \cap \ker Q^\dagger \subset \mathcal{H}^p$, and $P_B = \sum_i |\psi_i\rangle\langle\psi_i|$ is the orthogonal projector onto $\mathcal{B}$.

Fortuity concerns what happens to a BPS state when the theory is *uplifted* $N \rightarrow N' = N + 1$: the couplings C_{ijk} are inherited and independent new couplings $C_{ij,N+1}$ are drawn. A BPS state of the smaller theory need not remain BPS. The *uplift decoder* measures how much of an old BPS multiplet survives inside the new BPS space after reading a fixed occupation pattern. Concretely, all objects live on the charge- p Fock space $\mathcal{H} = \Lambda^p(\mathbb{C}^{N'})$ ($\dim \mathcal{H} = D$), and are built from two orthogonal projectors on it: P_B , onto the BPS space $\mathcal{B} = \mathcal{B}_{N'}^p$ ($\dim = d$), and $\Pi_T = 1 - n_{N'}$, onto

the occupation sector (the “slot”) $\mathcal{V}$ ($\dim = m$). The *uplift decoder* and its two Gram operators are

$$D_T = \Pi_T P_B, \quad K_T = D_T D_T^\dagger = \Pi_T P_B \Pi_T, \quad O_T = D_T^\dagger D_T = P_B \Pi_T P_B. \quad (2)$$

All three are $D \times D$ operators on $\mathcal{H}$: K_T is supported on the slot $\mathcal{V}$ (rank m), O_T on the BPS space $\mathcal{B}$ (rank d), and they share the same nonzero spectrum. We work throughout with $O_T = P_B \Pi_T P_B$ —the slot projector Π_T compressed between the BPS projectors, an LMRS-type “compressed projector” with $0 \preceq O_T \preceq 1$ —whose eigenvalues $\lambda \in [0, 1]$ are the squared cosines of the principal angles between $\mathcal{B}$ and $\mathcal{V}$: $\lambda = 1$ is a perfectly transmitted (fortuitous) direction, $\lambda = 0$ a lost one, and the interior $0 < \lambda < 1$ is fragile (metric-dependent) content. Reduced to an orthonormal BPS basis O_T is the $d \times d$ matrix we diagonalize (its slot-side twin K_T carries the same nonzero eigenvalues). The *uplift fidelity* $F_T = \frac{1}{d} \text{Tr } O_T = m_1$ and the fragility cost $\kappa_T = m_{-1} - 1 = \int \lambda^{-1} d\mu - 1$ are the low moments of its spectral measure $d\mu$.

1.2 Microscopic form of the decoder from the supercharge

The decoder admits an exact self-energy form directly in terms of the supercharge. Let Q_N be the old supercharge, $H_N = \{Q_N, Q_N^\dagger\}$ its Hamiltonian, and $G = H_N^{-1}$ on $(\ker H_N)^\perp$ (zero on $\ker H_N$) the Hodge Green’s operator, so that the old BPS projector is

$$P_B = 1 - Q_N G Q_N^\dagger - Q_N^\dagger G Q_N. \quad (3)$$

Uplift $N \rightarrow N' = N + 1$ and fix the enlarged charge p . Every state splits by the occupation of the new mode $e \equiv e_{N'}$,

$$|\Psi\rangle = |a\rangle + |b\rangle \wedge e, \quad |a\rangle \in \Lambda^p(\mathbb{C}^N) \text{ (empty)}, \quad |b\rangle \in \Lambda^{p-1}(\mathbb{C}^N) \text{ (occupied)}, \quad (4)$$

so that $\Pi_T = 1 - n_{N'}$ reads off the empty part, $\Pi_T |\Psi\rangle = |a\rangle$ and $\langle \Psi | \Pi_T | \Psi \rangle = \|a\|^2$. The uplifted supercharge splits accordingly, $Q_{N'} = Q_N + C'$, with $C' = (D \wedge e) \wedge (\cdot)$ the injection built from the *new* couplings $D = \sum_{i < j \leq N} C_{ij, N'} e_i \wedge e_j$ (the part of $Q_{N'}$ that moves the added mode). Sorting $Q_{N'} |\Psi\rangle$ and $Q_{N'}^\dagger |\Psi\rangle$ by the presence of e (with $e \wedge e = 0$) and demanding both vanish gives four conditions,

$$Q_N |a\rangle = 0, \quad Q_N^\dagger |b\rangle = 0, \quad Q_N |b\rangle = -D \wedge |a\rangle, \quad Q_N^\dagger |a\rangle = -\iota_D |b\rangle. \quad (5)$$

The first two make $|a\rangle$ closed and $|b\rangle$ co-closed; the last two couple them through the new couplings.

We eliminate $|b\rangle$. Writing $u \equiv D \wedge a = C' a \in \Lambda^{p+2}(\mathbb{C}^N)$, conditions (5)(ii)–(iii) read $Q_N^\dagger b = 0$ and $Q_N b = -u$, so $H_N b = Q_N^\dagger Q_N b = -Q_N^\dagger u$ and hence

$$|b\rangle = -G Q_N^\dagger (D \wedge a). \quad (6)$$

The occupied weight then collapses to a *single* resolvent insertion. Using $GH_N = 1$ on the (co-closed, non-harmonic) $|b\rangle$, then $H_N |b\rangle = Q_N^\dagger Q_N |b\rangle$, then $[G, Q_N] = 0$ and $u = -Q_N b$,

$$\|b\|^2 = \langle b | GH_N | b \rangle = \langle Q_N b | G | Q_N b \rangle = \langle a | C'^\dagger G C' | a \rangle, \quad (7)$$

an exact identity: the occupied norm is the new couplings propagated through the old resolvent. Here $C' = D \wedge (\cdot) : \Lambda^p \rightarrow \Lambda^{p+2}$ and G acts at the intermediate degree $p + 2$.

The decoder eigenvalue is the empty fraction, so (7) gives

$$\lambda = \frac{\langle \Psi | \Pi_T | \Psi \rangle}{\| \Psi \|^2} = \frac{\|a\|^2}{\|a\|^2 + \|b\|^2} = \frac{\|a\|^2}{\|a\|^2 + \langle a | C'^\dagger G C' | a \rangle}.$$

Parametrizing the BPS space by its closed empty components $|a\rangle$ —the physical norm is $\|\Psi\|^2 = \langle a|(1 + C'^\dagger G C')|a\rangle$ while the decoder measures $\|a\|^2$ —the eigenproblem $O_T|\Psi\rangle = \lambda|\Psi\rangle$ becomes $|a\rangle = \lambda(1 + C'^\dagger G C')|a\rangle$, i.e. the exact self-energy form

$$\boxed{O_T = (1 + C'^\dagger G C')^{-1} \quad \text{on } \ker Q_N \subset \Lambda^p(\mathbb{C}^N)} \quad (8)$$

with G the old resolvent at degree $p + 2$. This is a Schur-complement statement: the new mode leaks the BPS state out of the BPS locus by the self-energy $C'^\dagger G C'$, and O_T resums that leakage. To leading order in the new couplings it is the familiar self-energy $O_T = 1 - C'^\dagger G C' + \dots$, with fidelity defect $1 - O_T = C'^\dagger G C'/(1 + C'^\dagger G C') \simeq C'^\dagger G C'$. Equation (8) is what makes the disorder average tractable— C' is Gaussian and independent of the inherited data—and it is the operator form of the state-by-state identity $\lambda = E/(E + \sigma_a^2)$ of §6 (with $\langle a|C'^\dagger G C'|a\rangle/\|a\|^2 = \sigma_a^2/E$). It is the microscopic object whose ensemble statistics we now identify.

1.3 A diagrammatic language

The self-energy form (8) invites a Feynman-diagram reading of the decoder (Fig. 1). The rules are minimal: an external *BPS leg* is on shell ($H_N|\psi\rangle = 0$); the internal line is the *off-BPS propagator* $G = 1/H_N$, which vanishes on the BPS locus so that only virtual excursions off it propagate; the new couplings enter through a single *vertex* $C' = D\wedge$ that knocks a BPS state off the locus by attaching the added mode; and the inherited couplings enter through the old supercharge vertex $Q = C\wedge$. Reading left to right, a state is kicked off the locus by C' , propagates by G , and is kicked back by $C'^\dagger$: this is the self-energy $\Sigma = C'^\dagger G C'$, and the decoder $O_T = (1 + \Sigma)^{-1}$ resums the chain of such blobs into a *dressed BPS propagator*,

$$O_T = 1 - \Sigma + \Sigma^2 - \dots, \quad 1 - O_T \simeq \Sigma \quad (\text{leakage}). \quad (9)$$

Even the BPS projector is of this form: (3) reads $P_B = 1 - Q G Q^\dagger - Q^\dagger G Q$, the identity minus two self-energy blobs built from the old vertex Q .

Two features are transparent in this language. First, the *soft tail is a resonance*: a small- λ state is a diagram whose internal G -line is itself near-BPS ($H_N \rightarrow 0$), so G —and hence Σ —diverges and $\lambda \rightarrow 0$; the tail of §6 is a resonant self-energy. Second, this is the *microscopic*, single-realization companion of the ribbon/genus expansion (§2 onward): averaging over the Gaussian new couplings Wick-contracts the C' vertices pairwise, $\langle C' C' \rangle$, and each contraction is a ribbon edge, so disorder-averaging these Feynman diagrams *is* the fatgraph sum that the gravity reproduces. The two tiers—Feynman diagrams before averaging, ribbons after—are related by that one contraction.

1.4 The disorder-averaged law: Wachter/MANOVA

Because C' is independent Gaussian and the BPS subspace is (numerically) Haar-typical inside the charge sector, the disorder-averaged spectrum of O_T is the compression of one projector by an independent random subspace: a *Jacobi (MANOVA)* ensemble. With

$$D = \binom{N+1}{p} \quad (\text{sector dim}), \quad d = \dim \mathcal{B} = \tilde{D}_p - \tilde{D}_{N+1-p-q} \quad (\text{index}), \quad m = \binom{N}{p}, \quad (10)$$

and the two ratios $a = d/D$, $b = m/D = (N+1-p)/(N+1)$, the interior density is the *Wachter law*

$$\rho_0(\lambda) = \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi a \lambda(1 - \lambda)}, \quad \lambda_{\pm} = (\sqrt{a(1-b)} \pm \sqrt{b(1-a)})^2, \quad (11)$$

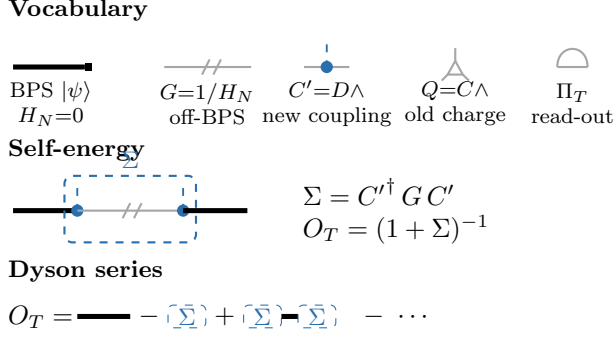


Figure 1: A Feynman–diagram language for the uplift decoder. *Top*: the vocabulary—BPS external legs (on shell, $H_N=0$), the off-BPS propagator $G=1/H_N$, the new-coupling vertex $C'=D\wedge$ (attaching the added mode, dashed), the inherited supercharge vertex $Q=C\wedge$, and the slot read-out Π_T . *Middle*: the self-energy insertion $\Sigma = C'^\dagger G C'$; the decoder is its resummation $O_T = (1 + \Sigma)^{-1}$. *Bottom*: equivalently O_T is the Dyson series (9), a dressed BPS propagator. Disorder-averaging Wick-contracts the C' vertices into the ribbon edges of §2; a near-BPS internal line ($H_N \rightarrow 0$) is the resonance that produces the soft tail.

supplemented by atoms $p_0 = \max(0, 1 - b/a)$ at $\lambda = 0$ and $p_1 = \max(0, 1 - (1 - b)/a)$ at $\lambda = 1$. Here $\tilde{D}_M(k) = \sum_n (-1)^n \binom{M}{k-qn}$ is the twisted counting function, and the closed form for a is the fortuity index itself.

Equivalently, ρ_0 is the free multiplicative convolution of two Bernoulli laws, $\text{Bern}(a) \boxtimes \text{Bern}(b)$ (Voiculescu freeness of the two projectors). Its S -transform factorizes, $S(z) = \frac{(1+z)^2}{(a+z)(b+z)}$, giving all moments in closed form:

$$m_1 = b, \quad m_2 = b(a + b - ab), \quad m_3 = b(a^2 + b^2 + 3ab - 3a^2b - 3ab^2 + 2a^2b^2), \quad \dots \quad (12)$$

with the full generating function fixed by the algebraic (genus-zero) spectral curve

$$(w - 1)z^2 + (w(a + b) - 1)z + wab = 0, \quad z = \psi(w) = \sum_k M_k w^k, \quad m_k = M_k/a. \quad (13)$$

Exact diagonalization confirms (11)–(12) in the bulk across $N = 6, \dots, 10$ (Kolmogorov–Smirnov distance $\lesssim 0.1$, tightening with N); the mean $\langle \lambda \rangle = b$ is an exact trace identity at every N (Figs. 2–3).

1.5 The protected part: a finite- N resonance

Two channels of the spectrum are *not* fragile and not captured by (11): exact atoms at $\lambda = 1$ (states in $\mathcal{B}_N^p \cap \ker C'$, transmitted exactly) and at $\lambda = 0$ (states in $\mathcal{B}_N^{p-1} \cap \ker C'^\dagger$, lost exactly), disorder-independent and in excess of the Wachter atoms p_0, p_1 . Their exact counts $n_1 = \dim(\mathcal{B}_N^p \cap \ker C')$ and $n_0 = \dim(\mathcal{B}_N^{p-1} \cap \ker C'^\dagger)$ are a *finite- N resonance*: they are appreciable only where $\dim \mathcal{B}_N^p \approx \binom{N}{p+2}$, near $N = 12$ – 13 , and vanish once the two separate. The total atom fraction $(n_0 + n_1)/d$ is 5.6% at $N = 12$ and $N = 13$ but *zero* at $N = 14, 15, 16$ (each over two disorder realizations): once $\dim \mathcal{B}_N^p \sim (\sqrt{3})^N$ drops below $\binom{N}{p+2} \sim 2^N/\sqrt{N}$ with a growing margin, C' acts injectively on $\mathcal{B}_N^p$ and no exact atoms survive. The protected core is thus a small transient, not a robust sector; the finite- N atoms dissolve into the near-edge tail as N grows, in keeping with the collapse of the band in §8.

Decoder spectrum: exact diagonalization vs analytic Wachter law (one-flavor, down channel, half filling)

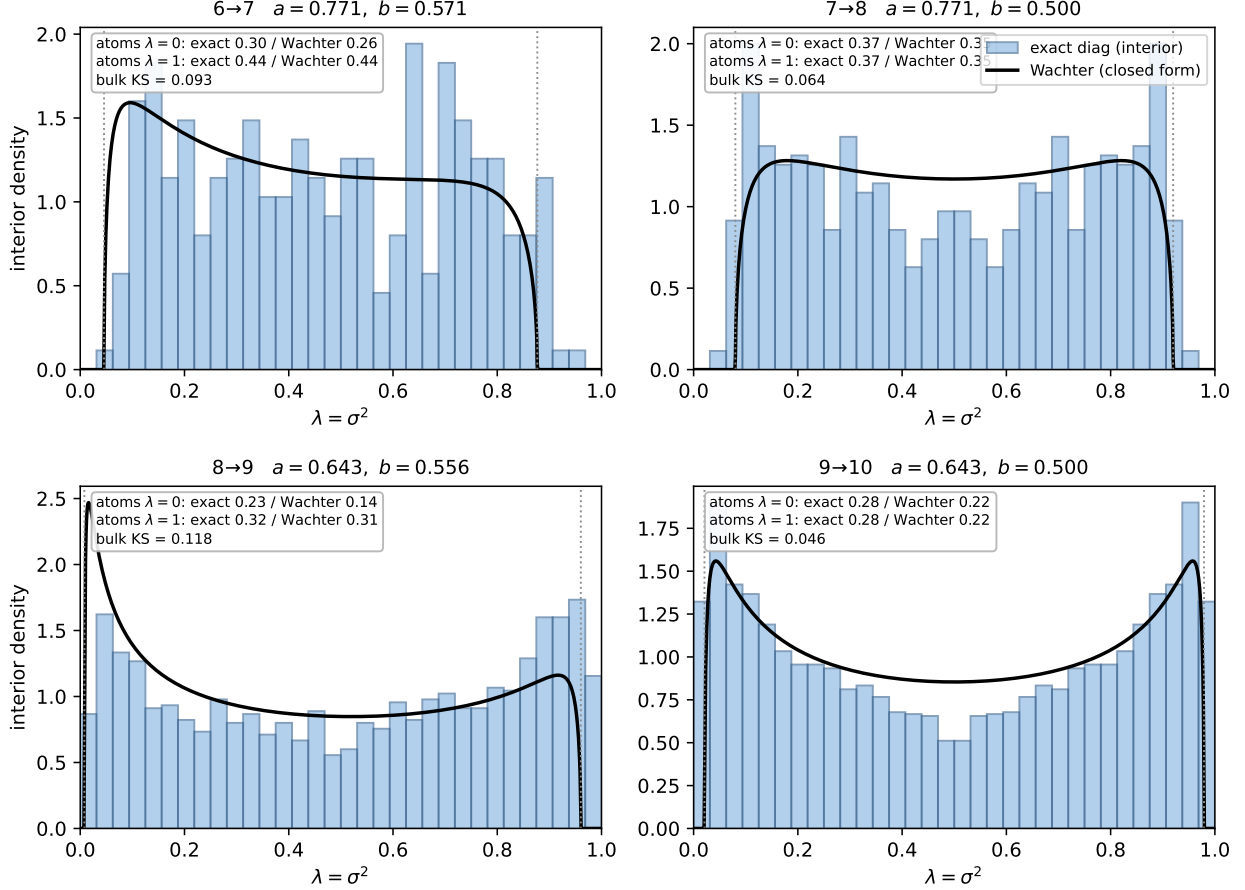


Figure 2: Interior decoder spectrum from exact diagonalization (blue histogram, pooled over disorder realizations) versus the analytic Wachter density (11) (black), for the one-flavor down channel at half filling, $N = 6, \dots, 9$. The parameters a, b are fixed by the closed-form dimensions (no fitting); the bulk Kolmogorov–Smirnov distance is $\lesssim 0.1$ and the endpoint atom weights track the Wachter predictions p_0, p_1 . The interior fills the band $[\lambda_-, \lambda_+]$ with the characteristic U -shape.

Distinct from these atoms is the $\mathbb{Z}_q$ -twisted Witten index, a robust object. The enlarged BPS dimension is $d = 3^{\lfloor N/2 \rfloor}$ (times 2 for odd N ; §8), so the BPS entropy density is exactly $\frac{1}{2} \ln 3$; in the two-flavor tower the protected BPS dimension obeys

$$\dim \mathcal{B}_{2m+1}^m = 3^m \iff S_0 = \frac{N-1}{2} \ln 3, \quad (14)$$

proved by a generating-function/Lagrange–Bürmann collapse. This index counts BPS states and sets the extremal entropy; it is *not* the decoder’s exact-atom count $n_0 + n_1$ above, which it exceeds by an order of magnitude already at $N = 12$. With that distinction the spectrum splits as

$$\underbrace{\rho_0(\lambda) \text{ (Wachter)}}_{\text{fragile / chaotic bulk}} + \underbrace{\text{a finite-}N \text{ atom resonance at } \lambda = 0, 1}_{\text{transient; 0 for } N \geq 14}, \quad (15)$$

and the object of this paper is a bulk (gravitational) account of the fragile part, with the protected atoms as its transient finite- N complement.

Exact-diagonalization decoder spectrum vs analytic Wachter law (interior ECDF vs CDF)

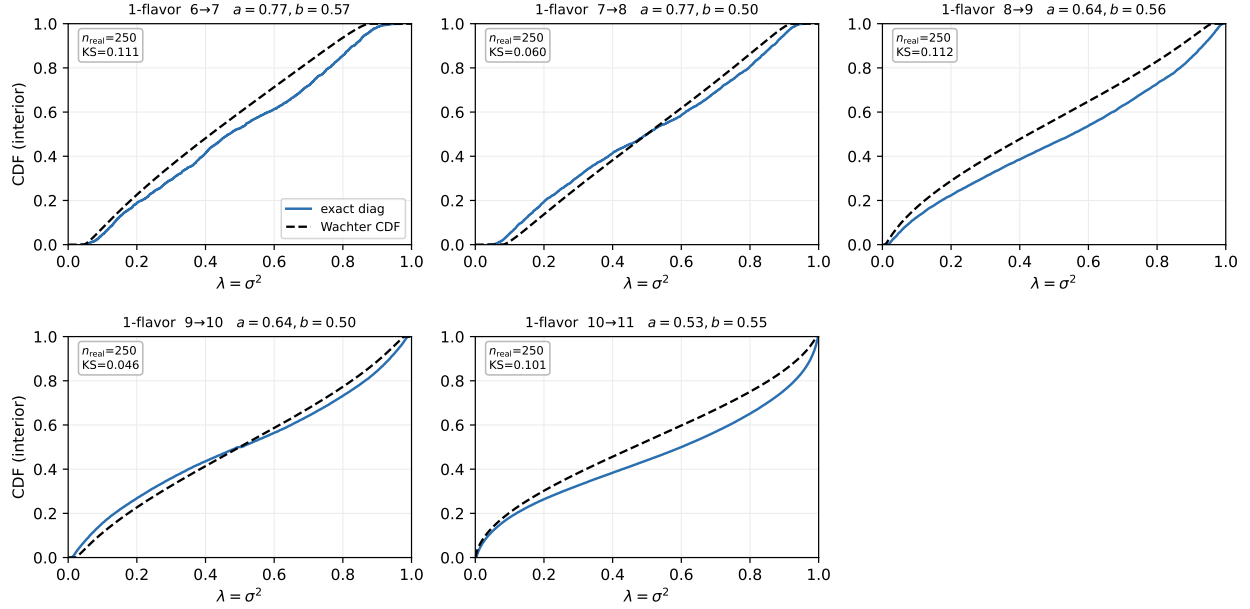


Figure 3: The same comparison as empirical CDF (blue) versus the Wachter CDF (dashed), cleaner at the edges, for the one-flavor down channel $N = 6 \rightarrow 7, \dots, 10 \rightarrow 11$. Agreement tightens with N (KS falling to ~ 0.05); the residual near $\lambda = 0, 1$ is the protected excess beyond the Wachter atoms.

2 The genus expansion of the decoder

The finite- D moments are exact and carry a topological grading. Writing $O_T = P_B \Pi_T P_B$ with $P_B = V V^\dagger$ for a Haar $D \times d$ isometry V , so that $\text{Tr } O_T^k = \text{Tr}(V^\dagger \Pi_T V)^k$, the Weingarten calculus gives, with $\gamma = (12 \dots k)$ the cyclic permutation and $c(\cdot)$ the number of cycles,

$$\langle \text{Tr } O_T^k \rangle = \sum_{\sigma, \tau \in S_k} \text{Wg}(\sigma \tau^{-1}; D) m^{c(\sigma)} d^{c(\gamma^{-1}\tau)}, \quad m_k = \frac{1}{d} \langle \text{Tr } O_T^k \rangle. \quad (16)$$

Each term is a ribbon graph whose Euler characteristic sets its power of $1/D$; expanding in $1/D = e^{-S_0}$ organizes the moment by genus,

$$m_k = m_k^{(0)}(a, b) + \frac{1}{D^2} m_k^{(1)}(a, b) + \frac{1}{D^4} m_k^{(2)}(a, b) + \dots \quad (17)$$

The genus-zero (planar) piece is exactly the Wachter law (12). The genus-one (single-handle) pieces are, in closed form,

$$m_1^{(1)} = 0, \quad (18)$$

$$m_2^{(1)} = -b(1-a)(1-b), \quad (19)$$

$$m_3^{(1)} = -b(1-a)(1-b)(10ab - 5a - 5b + 1), \quad (20)$$

$$m_4^{(1)} = -5b(1-a)(1-b)(14a^2b^2 - 14a^2b + 3a^2 - 14ab^2 + 10ab - a + 3b^2 - b). \quad (21)$$

The first moment receives no handle— $m_1 = b$ exactly, all orders—and every nonzero handle carries the common prefactor $b(1-a)(1-b)$, which vanishes when the subspace or slot fills the sector (a or $b \rightarrow 1$): no room for topology. Equations (19)–(21) are the targets the gravity side must reproduce. They have been checked against Haar Monte Carlo: e.g. at $a = b = \frac{1}{2}$, $m_2 = \frac{3}{8} - \frac{1}{4}D^{-2}$, $m_3 = \frac{5}{16} - \frac{3}{16}D^{-2}$, and $m_4 = \frac{35}{128} - \frac{15}{64}D^{-2}$, all matching to $O(D^{-4})$. The alternating signs and shared prefactor already hint that the whole tower resums to a single object, which we exhibit in §3.6.

2.1 Chord diagrams and the Narayana count

The same expansion has a gravity-native reading as a sum over bulk worldlines—*chord diagrams*—on the disk, the technology used for near-BPS operator spectra and Berry curvature in super-JT [13, 14]. Each factor of Π_T in $m_k = \frac{1}{a} \tau((P_B \Pi_T)^k)$ is a boundary insertion; the disk amplitude sums over bulk chords joining them, planar (non-crossing) chords being genus zero and crossings the handles, suppressed by $e^{-2S_0} = 1/D^2$ —exactly the grading (17).

Where the planar limit comes from. In the operator computations of [13, 14] the non-crossing (free) regime is reached by sending the inserted operator’s dimension $\Delta \rightarrow \infty$, which exponentially suppresses crossed worldlines. We have no such Δ ; its role is played by the large-sector-entropy planar limit $D = e^{S_0} \rightarrow \infty$, equivalently the Haar-typicality of the BPS subspace (§1). There the two projectors P_B and Π_T are asymptotically *free* (Voiculescu), so non-crossing diagrams dominate and crossings are the finite- D handles. The soft tail (§5) is exactly where this freeness fails—the BPS subspace is not Haar-typical along the near-BPS directions—so it lies outside the chord expansion, matching §6.

Catalan vs. Narayana. The nature of the insertion is decisive. A heavy *Gaussian* operator $\hat{\Lambda} = P_B \Lambda P_B$ contracts pairwise, so its non-crossing diagrams are non-crossing *pairings*, counted by the Catalan numbers $C_k = \frac{1}{k+1} \binom{2k}{k}$ and giving the *semicircle* [13, 14]. Our insertion Π_T is a *projector*, $\Pi_T^2 = \Pi_T$: a chord may enclose an entire run of insertions at no cost, so the non-crossing diagrams are non-crossing *partitions*, counted by the Narayana numbers

$$\text{Nar}(k, c) = \frac{1}{k} \binom{k}{c} \binom{k}{c-1}, \quad \sum_{c=1}^k \text{Nar}(k, c) = C_k, \quad (22)$$

the refinement of the Catalan count by the number of blocks c (Fig. 6). Weighting each partition by the free cumulants of the two Bernoulli spectra $\text{Bern}(a), \text{Bern}(b)$ and summing gives their free multiplicative convolution—the Wachter moments (12)—while the reference-block self-consistency (a chosen block cuts the disk into independent sub-disks) reproduces the quadratic spectral curve (13). The two-hard-edge Wachter law is thus the Narayana (projector) counterpart of the single-edge semicircle (Catalan/Gaussian), and reinstating crossings restores the handle tower (19)–(21).

The moments, explicitly. For two free projectors the disk moments are the non-crossing-partition sum

$$m_k = \frac{1}{a} \sum_{\pi \in \text{NC}(k)} \left(\prod_{V \in \pi} \kappa_{|V|} \right) b^{|K(\pi)|}, \quad \kappa_1 = a, \quad \kappa_2 = a(1-a), \quad \kappa_3 = a(1-a)(1-2a), \dots \quad (23)$$

the κ_s being the free cumulants of $\text{Bern}(a)$ and $K(\pi)$ the Kreweras complement ($|K(\pi)| = k+1-|\pi|$): each block of π is a ψ -index loop weighted by its cumulant, each block of $K(\pi)$ a t -index loop weighted by b . This is the planar necklace (27)—a non-crossing partition is a planar ribbon

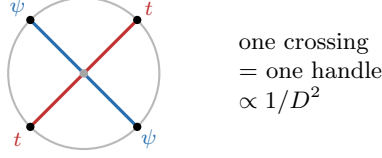


Figure 4: The genus-one handle as a crossing chord diagram ($k = 2$): the ψ -chord (blue) and t -chord (red) cross (gray dot), and each crossing is a handle weighting $1/D^2$. Summed at each k with the subleading Weingarten, the crossing diagrams give the handle tower (19)–(21) and hence (24) at $k = 2$.

graph—with $\kappa_2 = a(1-a)$ packaging the two-disk and planar-wormhole surfaces of §3 into one block. Enumerating the partitions (Fig. 5),

$$\begin{aligned} m_1 &= \frac{1}{a} \kappa_1 b = b, \\ m_2 &= \frac{1}{a} (\kappa_1^2 b + \kappa_2 b^2) = ab + (1-a)b^2 = b(a + b - ab), \\ m_3 &= \frac{1}{a} (\kappa_1^3 b + 3\kappa_2 \kappa_1 b^2 + \kappa_3 b^3) = b(a^2 + b^2 + 3ab - 3a^2 b - 3ab^2 + 2a^2 b^2), \end{aligned}$$

reproducing (12) term by term—the same disk values the gravitational sum gives in §3 (the factor 3 in m_3 is the three two-block partitions $\{12\}\{3\}$, $\{23\}\{1\}$, $\{13\}\{2\}$).

The handles: one crossing. Reinstating crossings promotes a non-crossing partition to a genus-one configuration, each crossing costing one handle, $1/D^2$. Already at $k = 2$ the exact finite- D moment is rational and shows both orders at once,

$$m_2 = \frac{b[D^2(a + b - ab) - 1]}{D^2 - 1} = \underbrace{b(a + b - ab)}_{\text{disk}} - \underbrace{b(1-a)(1-b)}_{m_2^{(1)}} \frac{1}{D^2} + O(D^{-4}), \quad (24)$$

so the single crossing diagram of Fig. 4—the wormhole whose ψ - and t -chords must cross, i.e. the handle—together with the $1/D^2$ dressings of the planar terms gives $m_2^{(1)} = -b(1-a)(1-b)$, matching (19). The same crossing sum at $k = 3$ gives $m_3^{(1)} = b(1-a)(1-b)(10ab - 5a - 5b + 1)$, and the once-crossing partitions, weighted by the subleading Weingarten, reproduce the entire handle tower (19)–(21)—which we have verified symbolically through m_4 . The chord diagrams thus generate *both* the disk (non-crossing, Narayana/Wachter) and the handles (crossing): an exact alternative to the surface sum of §3.

Relation to the Berry-curvature computation. Reference [14] applies this same chord technology in $\mathcal{N}=2$ super-JT to a different overlap observable: the Berry curvature $F_{\mu\nu} = i[\hat{\Lambda}_\mu, \hat{\Lambda}_\nu]$ of the moduli-space connection, a commutator of two projected *Gaussian* operators, whose spectrum is the free commutator of two semicircles. Our decoder $O_T = P_B \Pi_T P_B$ is a compression of two *projectors*, giving the two-edge Wachter/Narayana law, and its ensemble breaks down through the macroscopic soft tail (§5), which has no counterpart in the edge-regular commutator spectrum. The two are complementary overlap-side chaos diagnostics for fortuitous BPS states, computed by the same super-JT machinery.

2.2 The cylinder and the spectral form factor

The disk gave the moments; the *cylinder*—two trace boundaries joined through the bulk—gives the connected double-trace moments $\langle \text{Tr } O_T^m \text{ Tr } O_T^n \rangle_c$ and hence the spectral form factor. In chord

$$\begin{aligned}
m_2 : & \quad \begin{array}{c} \text{circle with 2 dots} \\ \kappa_1^2 b \end{array} + \begin{array}{c} \text{circle with 2 dots and a blue chord} \\ \kappa_2 b^2 \end{array} = b(a+b-ab) \\
m_3 : & \quad \begin{array}{c} \text{circle with 3 dots} \\ \kappa_1^3 b \end{array} + \begin{array}{c} \text{circle with 3 dots and a blue chord} \\ \kappa_2 \kappa_1 b^2 \end{array} + \begin{array}{c} \text{circle with 3 dots and a blue chord} \\ \kappa_2 \kappa_1 b^2 \end{array} + \begin{array}{c} \text{circle with 3 dots and a blue chord} \\ \kappa_2 \kappa_1 b^2 \end{array} + \begin{array}{c} \text{circle with 3 dots and a blue triangle} \\ \kappa_3 b^3 \end{array}
\end{aligned}$$

Figure 5: The non-crossing partitions generating the disk moments via (23). Blue chords group the ψ -insertions into blocks (a block of size s weights κ_s ; singletons carry no chord), and the complementary t -loops weight b each. Summing the two diagrams for m_2 and the five for m_3 gives the Wachter moments (12); crossings (not shown) add the $1/D^2$ handles.

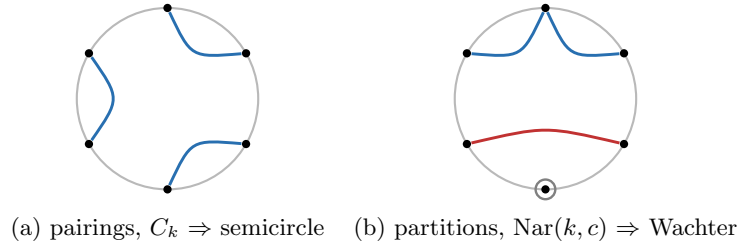
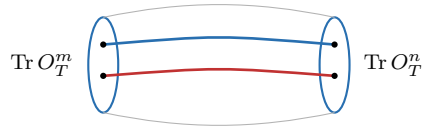


Figure 6: Disk chord diagrams for the decoder moments. (a) A Gaussian insertion contracts pairwise, so the planar diagrams are non-crossing *pairings* (Catalan C_k)—the semicircle case of [13, 14]. (b) The projector Π_T ($\Pi_T^2 = \Pi_T$) lets one chord enclose a whole block, so the planar diagrams are non-crossing *partitions* (Narayana $\text{Nar}(k, c)$; here the blocks $\{1, 2, 6\}$ blue, $\{3, 5\}$ red, and the singleton $\{4\}$), giving the two-hard-edge Wachter law. Crossings, not drawn, are the $1/D^2$ handles.

language these are annular non-crossing partitions on the two boundaries [14]; equivalently they are the universal two-point resolvent (the “double trumpet”) of the Wachter spectral curve,

$$W_{0,2}(x_1, x_2) = \frac{1}{2(x_1 - x_2)^2} \left[\frac{x_1 x_2 - \frac{\lambda_- + \lambda_+}{2}(x_1 + x_2) + \lambda_- \lambda_+}{\sigma(x_1) \sigma(x_2)} - 1 \right], \quad \sigma(x) = \sqrt{(x - \lambda_-)(x - \lambda_+)}. \quad (25)$$



This $W_{0,2}$ is *universal*—fixed by the two hard edges $\lambda_{\pm}$ alone, independent of the potential; it is the genus-zero, two-boundary object of the same topological recursion that produced W_1 in App. 3.6. Its double discontinuity across the cut is the connected density correlator, whose smooth part is the universal sine kernel $\langle \rho(\lambda) \rho(\lambda') \rangle_c \sim -1/[2\pi^2(\lambda - \lambda')^2]$, which Fourier transforms to a linear ramp. With the auxiliary-time form factor $\text{SFF}_c(t) = \langle \text{Tr } e^{iO_T t} \text{Tr } e^{-iO_T t} \rangle_c$, the cylinder gives the Gaussian-unitary ramp and plateau,

$$\text{SFF}_c(t) \simeq \frac{t}{2\pi} \quad (\text{ramp}), \quad \text{SFF}_c \rightarrow d \quad \text{at the Heisenberg time } t_H = 2\pi d \quad (\text{plateau}). \quad (26)$$

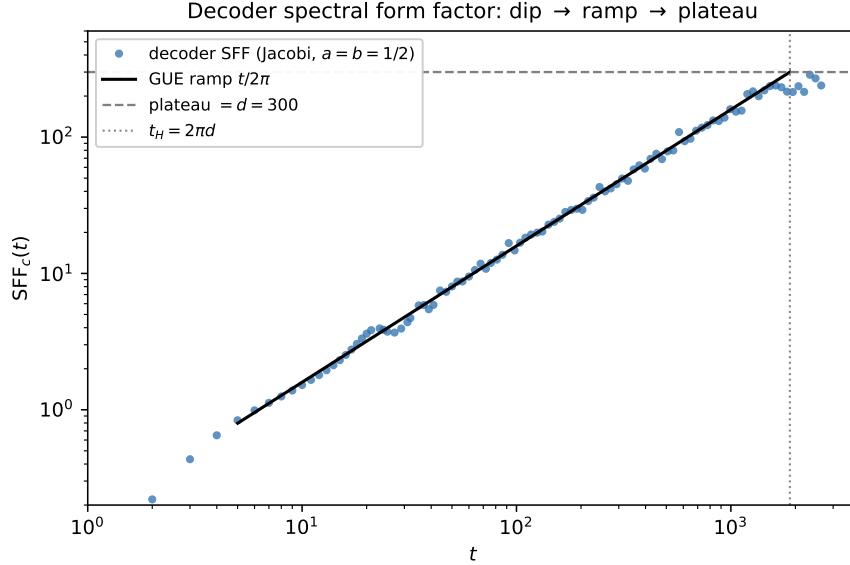


Figure 7: The decoder spectral form factor from exact diagonalization of the Jacobi ensemble ($a=b=\frac{1}{2}$, $d = 300$, pooled over 240 realizations): dip, then the Gaussian-unitary ramp $t/2\pi$ (black line), then the plateau $\text{SFF}_c = d$ at the Heisenberg time $t_H = 2\pi d$. The overlap spectrum is random-matrix correlated throughout its band—the smooth-correlator counterpart of the adjacent-gap result $\langle r \rangle = \text{GUE}$.

Exact diagonalization of the Jacobi ensemble confirms this over two decades (Fig. 7): an initial dip, a ramp of fitted slope $0.159 \approx 1/2\pi$, then the plateau at d . This is the *smooth-correlator* face of the chaos already seen discretely in $\langle r \rangle = \text{GUE}$ (§9): the decoder’s overlap spectrum carries random-matrix correlations throughout its Wachter band. The ramp is the same super-JT cylinder that produces the energy-side and Berry-curvature [14] form factors; only the boundary observable differs (two projectors, hence the Wachter curve and its two edges in (25)). It is unaffected by the soft tail—a disk (density) feature, whereas the ramp is the connected cylinder—consistent with the tail being non-universal in density but GUE in correlations (§9).

3 The gravitational model

3.1 Two brane species in the topological sector

In the West-Coast / PSSY model [1] JT gravity is decorated with EOW branes whose flavor index labels black-hole microstates; the disorder-averaged Gram matrix of the microstates is a random matrix, and connected *replica wormholes* generate its off-diagonal statistics, so that its spectrum is Marchenko-Pastur and its rank is bounded by e^{S_0} [2, 3]. We use the same machinery with *two* brane species:

- ψ -branes carrying the d BPS flavors ($|\psi_i\rangle$, $i = 1, \dots, d$);
- t -branes carrying the m flavors of the slot, $\Pi_T = \sum_{\alpha=1}^m |t_\alpha\rangle\langle t_\alpha|$.

A decoder moment is then a *necklace* of gravitational overlaps,

$$\text{Tr}(P_B \Pi_T)^k = \sum_{\{i_\alpha\}} \sum_{\{\alpha_\alpha\}} \langle \psi_{i_1} | t_{\alpha_1} \rangle \langle t_{\alpha_1} | \psi_{i_2} \rangle \langle \psi_{i_2} | t_{\alpha_2} \rangle \cdots \langle t_{\alpha_k} | \psi_{i_1} \rangle, \quad (27)$$

and the gravitational path integral instructs us to sum over all ways bulk geometry glues the $2k$ overlap edges. Because the BPS states sit at $E = 0$, the $\mathcal{N}=2$ super-Schwarzian is frozen onto its index $\delta(E)$: there is *no propagating energy*, only the topological brane combinatorics. This is precisely why the dual is one of *overlaps*, not geometry.

3.2 Amplitudes and Feynman rules

The topological sector supplies a minimal set of amplitudes:

<i>disk</i>	$Z_1 = e^{S_0} = D$ (sector degeneracy)
<i>wormhole tube</i>	weight $\frac{1}{D^2 - 1} = \frac{1}{D^2} \sum_{h \geq 0} D^{-2h}$ (one handle per factor e^{-2S_0})
ψ -index loop	factor $d = aD$
t -index loop	factor $m = bD$
<i>genus-g surface</i>	net suppression D^{-2g}

The exact Weingarten kernel is the all-handle resummation of the wormhole tube, so (16) is the gravitational sum over topologies for (27).

3.3 Why the disk is Wachter: the 't Hooft mechanism

A single connected wormhole costs $1/D^2$, but the ψ - and t -flavor loops threading it pay $md \sim abD^2$, exactly compensating. Hence *all planar wormholes are unsuppressed* and must be resummed; the resummation is a Schwinger–Dyson equation identical to the free subordination equation for $\text{Bern}(a) \boxtimes \text{Bern}(b)$ —the algebraic curve (13)—and its solution is the Wachter density (11). PSSY’s Marchenko–Pastur law is the dilute limit $b \rightarrow 0$ (a rare slot = few branes) of this two-parameter law. Only geometries with an *uncompensated* handle survive at order $1/D^2$.

3.4 The four bulk surfaces of m_2

For $k = 2$ the sum over gluings yields exactly four ribbon graphs, Fig. 8. Three are planar and one carries a handle:

(σ, τ)	topology	genus	weight
(e, e)	two disks (disconnected)	0	b^2
(e, swap)	planar ψ -wormhole + t -loop	0	$-ab^2$
$(\text{swap}, \text{swap})$	connected wormhole	0	ab
(swap, e)	handle	1	$-b/D^2$

The genus-zero surfaces sum to $b^2 - ab^2 + ab = b(a + b - ab) = m_2^{(0)}$, the Wachter disk. The handle, together with the $O(1/D^2)$ curvature dressings of the three planar surfaces, sums to

$$m_2^{(1)} = -b(1 - a)(1 - b), \quad (28)$$

matching (19) exactly. This is the gravity side producing the first handle from an honest sum over topologies.

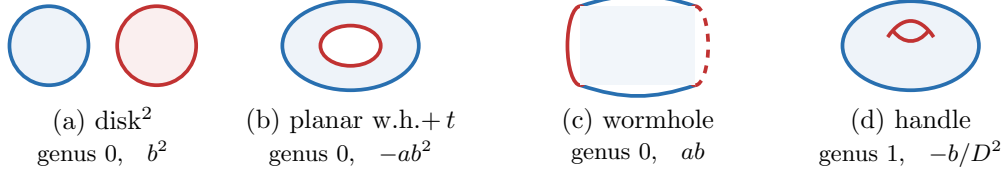


Figure 8: The four bulk surfaces contributing to m_2 . Blue (ψ) and red (t) mark the two brane species. Panels (a)–(c) are planar (genus 0) and resum to the Wachter disk $b(a + b - ab)$; the flavor loops on the connected wormhole (c) compensate its $1/D^2$ cost, so it is unsuppressed. Panel (d) is the single uncompensated **handle**; with the curvature dressings of (a)–(c) it yields $m_2^{(1)} = -b(1 - a)(1 - b)$.

3.5 m_3 : the same machine, next order

At $k = 3$ the gluing sum produces 12 planar surfaces, 21 leading genus-one surfaces, and 3 genus-two surfaces. Grouped by topology,

$$m_3^{(0)} = b(a^2 + b^2 + 3ab - 3a^2b - 3ab^2 + 2a^2b^2) \quad (\text{disk, Wachter}), \quad (29)$$

$$m_3^{(1)} = b(1 - a)(1 - b)(10ab - 5a - 5b + 1) \quad (\text{handle}), \quad (30)$$

both reproduced exactly by the topological expansion, in agreement with (20). The leading coefficient of m_k is the Catalan number C_{k-1} (the Fuss–Catalan fingerprint of freeness), providing an independent check that the disk is the free convolution. The check extends to m_4 (agreeing with (21), verified against Haar data), with the disk keeping its Catalan leading coefficient $C_3 = 5$.

3.6 The genus-one resolvent in closed form

The whole handle tower resums to a single meromorphic form. Applying Eynard–Orantin topological recursion [5] to the Wachter spectral curve—uniformized by Zhukovsky $x(z) = \alpha + \gamma(z + z^{-1})$ with branch points at $z = \pm 1$, disk data $y(z) - y(1/z) = -\sigma(x)/(ax(1 - x))$, and $\gamma^2 = ab(1 - a)(1 - b)$, $\alpha = a + b - 2ab$ —the genus-one one-point function $\omega_{1,1}$ (spelled out in Appendix A) evaluates to

$$W_1(x) = -\frac{b(1 - a)(1 - b)x(x - 1)}{[(x - \lambda_-)(x - \lambda_+)]^{5/2}} = -\frac{\gamma^2 x(x - 1)}{a\sigma(x)^5} \quad (31)$$

where $W_1(x) = \sum_{k \geq 2} m_k^{(1)}/x^{k+1}$ and $\sigma(x) = \sqrt{(x - \lambda_-)(x - \lambda_+)}$. The single normalization subtlety—the $\frac{1}{D} \text{Tr}$ versus $\frac{1}{d} \text{Tr}$ bookkeeping between the $D \times D$ matrix and the $d \times d$ compression—enters as an overall factor $-1/a^2$, fixed exactly at six independent (a, b) points. Expanding (31) at large x regenerates (19)–(21) identically, and predicts

$$m_5^{(1)} = 5b(1 - a)(1 - b)(84a^3b^3 - 126a^3b^2 + 56a^3b - 7a^3 - 126a^2b^3 + 154a^2b^2 - 49a^2b + 3a^2 + 56ab^3 - 49ab^2 + 8ab - 7b^3 + 3b^2). \quad (32)$$

Three features confirm (31) is the genuine genus-one object rather than a fit: (i) its only singularities are at the branch points $\lambda_{\pm}$; in the local coordinate ζ with $x - \lambda_{\pm} \sim \zeta^2$, the differential $W_1 dx \sim \zeta^{-4} d\zeta$ is a pole of order $6g - 4 + 2n = 4$ with no residue—the universal structure of $\omega_{1,1}$. (ii) The numerator $x(x - 1)$ vanishes at the hard walls $\lambda = 0, 1$, so the handle contributes *nothing*

at the protected–atom locations: the wormhole tower never touches the fortuity core, now visible in the analytic structure of W_1 itself. (iii) The common prefactor $b(1-a)(1-b)$ of all handle moments is manifest as the overall coefficient, and the resolvent’s genus–one correction is purely odd across the cut (no rational part), so a single line encodes the entire alternating–sign tower.

4 The amplitude dictionary from the super–JT action

So far the amplitudes of §3 were *adopted* from the matrix–integral duality. We now argue that in the BPS sector they instead *follow* from the $\mathcal{N}=2$ super–JT action, and that this simultaneously settles the all–genus statement.

4.1 The action and its topological term

The $\mathcal{N}=2$ super–JT action for a surface Σ is

$$I = \underbrace{-S_0 \chi(\Sigma)}_{\text{topological}} + I_{\text{super-Sch}}[\partial\Sigma] + I_{\text{brane}}[\text{EOW}], \quad -S_0\chi = -\frac{S_0}{2\pi} \left(\frac{1}{2} \int_{\Sigma} \sqrt{g} R + \int_{\partial\Sigma} \sqrt{h} K \right), \quad (33)$$

the last equality being Gauss–Bonnet. The topological term alone weights every connected surface by

$$e^{-I_{\text{top}}} = e^{S_0\chi(\Sigma)}. \quad (34)$$

This is exact and requires no dynamical input: it is simply the value of the Einstein–Hilbert term in two dimensions.

4.2 Why the dressing trivializes: the decoder is kinematic

In general JT the full amplitude is (34) *dressed* by the (super–)Schwarzian path integral—the disk is $e^{S_0} Z_{\text{Sch}}(\beta)$, the cylinder is a nontrivial double–trumpet, and so on. The dressing carries all the finite–energy dynamics. The decoder, however, is built *entirely from zero–energy states*: P_B projects onto the BPS subspace, and $O_T = P_B \Pi_T P_B$ is a purely kinematic operator on it. A gravitational quantity that depends only on the BPS Hilbert space cannot depend on the Schwarzian dynamics that governs finite E ; it can only see the topological sector. Concretely, the super–Schwarzian density of states is $\rho(E) = e^{S_0} \delta(E) + \rho_{\text{cont}}(E)$ with a gap [6, 7, 8]; projecting each boundary to $E = 0$ keeps only the $\delta(E)$, whose weight is the degeneracy e^{S_0} . Thus the dressing on every BPS–projected boundary collapses to the number e^{S_0} , and the amplitude of a surface reduces to its topological value (34):

$$Z(\Sigma)|_{\text{BPS}} = e^{S_0\chi(\Sigma)} \times (\text{brane flavor traces}). \quad (35)$$

At the disk this is the exact BPS degeneracy $Z_1 = e^{S_0}$ [6, 8]; the statement for higher topology is the kinematic (energy–independent) extension, consistent with the super–JT matrix–model structure [7].

4.3 Branes and the assembly

The two projectors enter as two EOW–brane species: ψ –branes (d BPS flavors) for P_B and t –branes (m flavors) for Π_T , exactly the microstate branes whose overlaps the gravitational path integral computes [8, 1]. A brane segment conserves its flavor index, so a closed ψ –loop contributes $\sum_{i=1}^d 1 =$

d and a closed t -loop m ; distinct species share no disk, connecting only through wormholes. With $e^{S_0} = D$ and $\chi = 2 - 2g - (\#\text{boundaries})$, Eq. (35) gives immediately

$$\text{disk } (\chi=1) : e^{S_0} = D, \quad \text{each handle} : e^{-2S_0} = D^{-2}, \quad \psi\text{-loop} : d, \quad t\text{-loop} : m, \quad (36)$$

which is precisely the amplitude dictionary of §3—now read off the action rather than imposed.

4.4 All genus, and the unification

The gravitational path integral sums (35) over all fillings of a moment’s brane necklace. By the standard ’t Hooft/fatgraph duality—the D -power of a ribbon graph equals the Euler characteristic of its dual surface—this sum is *identically* the Weingarten average (16). Hence the action reproduces the entire genus expansion: the disk is the Wachter law, genus one is the closed form W_1 of (31), and every handle follows. The three descriptions coincide,

$$\boxed{\text{super-JT action } e^{S_0\chi} \iff \text{Weingarten genus expansion} \iff \text{topological recursion on (13)}} \quad (37)$$

and “the action reproduces the handles to all orders” holds because the BPS sector is *topological*: there is no separate dynamical wormhole to evaluate. The handle coefficients are purely combinatorial (fatgraph counts)—which is exactly why they were computable from Weingarten and resumable into W_1 —while the action’s sole contribution, the weight $e^{S_0\chi}$, is exact.

Status of the derivation

Derived (exact): the weight $e^{S_0\chi}$ from the topological term; the flavor traces from the brane boundary conditions; hence the dictionary, the genus counting, and—via the fatgraph duality—the all-genus reproduction of the Wachter law and W_1 . *Derived in the literature*: the disk degeneracy $Z_1 = e^{S_0}$ and the BPS-microstate overlap/Gram-matrix construction [6, 8, 7]. *Assumed*: that BPS localization trivializes the super-Schwarzian dressing at all genus (proved at the disk; argued here from the kinematic nature of the decoder and the super-JT matrix-model structure), and the sum-over-topologies as the gravitational path integral. *Genuinely open*: the extension *off* the BPS point, $E \neq 0$, where the Schwarzian dynamics no longer trivializes and reproducing the amplitudes would require a true dynamical super-JT computation. At $E = 0$ the derivation is complete; away from it lies the frontier.

5 Where the ensemble fails: the macroscopic soft tail

The gravitational/ensemble description of §3–4 reproduces every positive moment $m_{k \geq 1}$ of the decoder. It does *not* reproduce the exact spectrum near $\lambda = 0$, and the discrepancy is not small.

A macroscopic, non-Haar tail. Exact diagonalization in the large- N ($a < b$) regime places a population of BPS states *below* the Wachter hard edge λ_- , extending down to $\lambda \sim 10^{-4}$ – 10^{-5} (Fig. 9, left). A Haar-random subspace at the identical (D, d, m) has *zero* sub-edge eigenvalues (checked at every N), so the tail is genuinely non-Haar: invisible to the free-probability law and to the wormhole sum built on it. Crucially it is *macroscopic*—an $O(10\%)$ fraction of all BPS states (9%, 9%, 15%, 14% at $N = 12$ –15; Fig. 9, right)—and it *grows* as $a = d/D$ decreases (i.e. with N), from $\simeq 9\%$ at $a = 0.42$ to $\simeq 15\%$ at $a = 0.34$. Two controls confirm this is the tail growing, not the edge sliding: the *deep*-tail fraction (states below $\lambda_-/5$, insensitive to small edge shifts) rises in step

(4% $\rightarrow$ 6%), while the Kolmogorov–Smirnov distance of the *bulk* $[\lambda_-, \lambda_+]$ to the Wachter law stays flat (0.08 $\rightarrow$ 0.07). So the bulk is universal and improving, and the tail is the non-universal excess, growing. This is a genuine breakdown of random-matrix universality for the fortuitous overlap spectrum, not a finite-size or dilute effect.

What the tail is. The decoder eigenvalue is exactly $\lambda = \langle \psi | (1 - n_{N'}) | \psi \rangle$, so the small- λ states are BPS states concentrated on the *newly added mode being occupied*—the “almost-lost” content. The tail is the structured, near-boundary part of the fortuity window that the generic (Haar) model cannot see.

The dressing cost is edge-sensitive, not a runaway. Because $\kappa = \int \lambda^{-1} d\mu - 1$ is dominated by the smallest λ , it is controlled by this tail. But κ is *not* a clean function of N : it peaks where the ensemble edge itself approaches zero (near the $a = b$ line) and is otherwise moderate ($\kappa = 115, 282, 44, 45$ at $N = 12\text{--}15$). The robust, universal diagnostic of the breakdown is the tail *fraction*, not κ ; the early “ κ runs away” reading was an artifact of sampling near criticality.

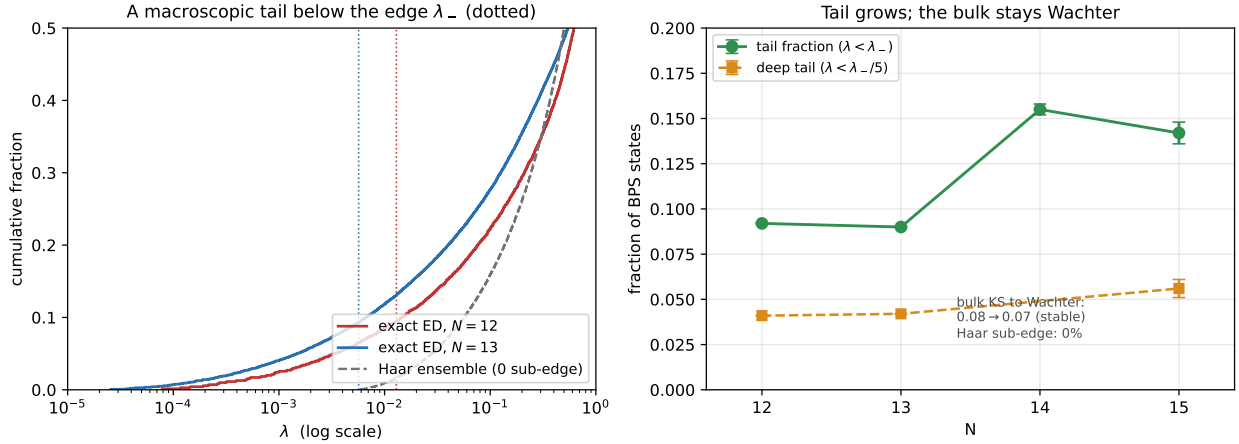


Figure 9: *Left*: exact decoder spectra ($N = 12, 13$) leak an $O(10\%)$ fraction of states below the Wachter edge λ_- (dotted), down to $\lambda \sim 10^{-5}$; a Haar-random subspace at the same (D, d, m) has no sub-edge weight (dashed). *Right*: both the tail fraction ($\lambda < \lambda_-$) and the deep-tail fraction ($\lambda < \lambda_-/5$) grow with N , while the bulk stays Wachter (KS 0.08 $\rightarrow$ 0.07) and the Haar sub-edge weight is 0% throughout. The growing tail is the part of the fortuitous overlap spectrum that the RMT/gravity ensemble misses.

Status. *Firm* (over $N = 12\text{--}15$): the tail exists, is non-Haar (Haar gives 0% at every N), is macroscopic, and *grows*—in both the total and the deep fraction—while the bulk remains Wachter. The breakdown is therefore not saturating over the accessible range. Its *origin* is identified (§6): the tail is the smaller theory’s near-BPS spectrum imaged by the uplift, via the exact identity $\lambda = E/(E + \sigma_a^2)$ derived from the BPS constraint (linear, $\lambda \simeq cE$, in the tail). *Open* (larger N): whether the tail fraction levels off at an $O(10\%)$ sector or continues to climb toward $O(1)$, in which case the ensemble/gravity would capture a *vanishing* fraction of the spectrum as $N \rightarrow \infty$. The small- λ density is *not* a fixed power law—the fitted exponent flattens steadily, $\rho \sim \lambda^{-0.4} \rightarrow \lambda^{-0.13}$ over $N = 12\text{--}15$ —consistent with its inheriting the drifting near-gap profile of ρ_H .

6 The gravitational origin of the soft tail

The tail is not structureless. Its origin can be derived exactly from the BPS constraint, and the derivation ties each tail state, one for one, to a near-BPS state of the *smaller* theory; the numerics then confirm the derived law to machine precision.

Everything follows from the block decomposition (4) and the four BPS conditions (5) of §1.2, now read off state by state at the two hard edges: with $|\Psi\rangle = |a\rangle + |b\rangle \wedge e_{N'}$ ($|a\rangle$ empty at degree P , $|b\rangle$ occupied at degree $P - 1$),

$$Q_N|a\rangle = 0, \quad Q_N^\dagger|b\rangle = 0, \quad Q_N|b\rangle = -D \wedge |a\rangle, \quad Q_N^\dagger|a\rangle = -\iota_{\bar{D}}|b\rangle.$$

The lower edge comes from the third of these. With $|b\rangle \in \ker Q_N^\dagger$ we have $\langle b|H_N|b\rangle = \|Q_N|b\rangle\|^2 \equiv E\|b\|^2$ under the smaller theory's Hamiltonian $H_N = \{Q_N, Q_N^\dagger\}$, so condition (5)(iii) is the *exact norm balance* $\|D \wedge a\|^2 = \|Q_N|b\rangle\|^2 = E\|b\|^2$. The decoder eigenvalue is $\lambda = \|a\|^2/(\|a\|^2 + \|b\|^2)$, and eliminating the norms gives the identity

$$\boxed{\lambda = \frac{E}{E + \sigma_a^2}}, \quad \sigma_a^2 \equiv \frac{\|D \wedge a\|^2}{\|a\|^2}, \quad E \equiv E_b = \frac{\langle b|H_N|b\rangle}{\|b\|^2}, \quad (38)$$

with E_b the energy of the occupied component. The same machinery gives a second identity for free. The Jacobi law has *two* hard edges, at $\lambda = 0$ and $\lambda = 1$; the third condition produced the lower, and its mirror, condition (5)(iv) $Q_N^\dagger a = -\iota_{\bar{D}}b$, produces the upper. Applied to a state whose empty component is near-BPS, $|a\rangle \in \ker Q_N$ so that $\|Q_N^\dagger a\|^2 = \langle a|H_N|a\rangle \equiv E_a\|a\|^2$, it gives $\|\iota_{\bar{D}}b\|^2 = E_a\|a\|^2$ and hence

$$1 - \lambda = \frac{E_a}{E_a + \sigma_b^2}, \quad \sigma_b^2 = \frac{\|\iota_{\bar{D}}b\|^2}{\|b\|^2}. \quad (39)$$

The two edges are thus governed by the same balance at complementary fermion numbers: the lower by near-BPS *occupied* components (degree $P - 1$, $E_b \rightarrow 0$), the upper by near-BPS *empty* components (degree P , $E_a \rightarrow 0$).

Both identities are exact for every state, but they simplify in the tails. There $E \ll \sigma_a^2$, and (38) linearizes to

$$\lambda \simeq \frac{E}{\sigma_a^2} = cE, \quad c = \frac{1}{\langle \sigma_a^2 \rangle}, \quad (40)$$

because σ_a^2 —the mean-square strength of the new-coupling wedge on the empty component—is a self-averaging Gaussian quantity that varies little across states while E sweeps the near-BPS spectrum. The soft-tail density is therefore a rescaled image of the near-BPS energy density of the smaller theory,

$$\rho_{\text{tail}}(\lambda) = \frac{1}{c} \rho_H(E = \lambda/c). \quad (41)$$

The self-energy form of the decoder reads off the same conclusion. In (8), $O_T = (1 + C'^\dagger G C')^{-1}$ with $G = 1/H_N$ the resolvent of the smaller theory, a decoder eigenvalue near zero is a self-energy near its maximum, sourced by the large eigenvalues of G —the *small* eigenvalues of H_N , i.e. the near-BPS states. The BPS grading above is the state-by-state version of this statement, and it is what turns the qualitative “near-BPS states source the tail” into the sharp linear map (40).

These predictions are borne out sharply. For each enlarged BPS eigenvector we extract its occupied component $|b\rangle \in \Lambda^{P-1}(\mathbb{C}^N)$ and its empty component $|a\rangle \in \Lambda^P(\mathbb{C}^N)$ and evaluate their energies under H_N . The eigenvalue tracks the occupied energy almost deterministically (Fig. 10,

left): the correlation is 0.93–0.95, the tail states sit at $\langle H_N \rangle \sim 0.05$ –0.2 against a bulk ~ 70 (a 10^3 separation), and the identity (38) holds to machine precision, with fitted slope $c \approx 0.018$ (0.96–0.98 of the predicted value at $N = 12, 13$). The dual identity (39) holds equally well at the upper edge (Fig. 10, right): at $N = 12$ it governs 31 near-BPS empty states at $E_a \sim 0.01$ against a bulk $E_a \sim 13$ (correlation 0.91), versus the 67 states of the lower tail, the two differing in size because the near-BPS densities at fermion numbers $P - 1$ and P differ. The two hard edges are thus two overlap shadows of the smaller theory’s near-BPS spectrum, at complementary fermion numbers—a structure with no analogue in the single-hard-edge (Bessel) energy model.

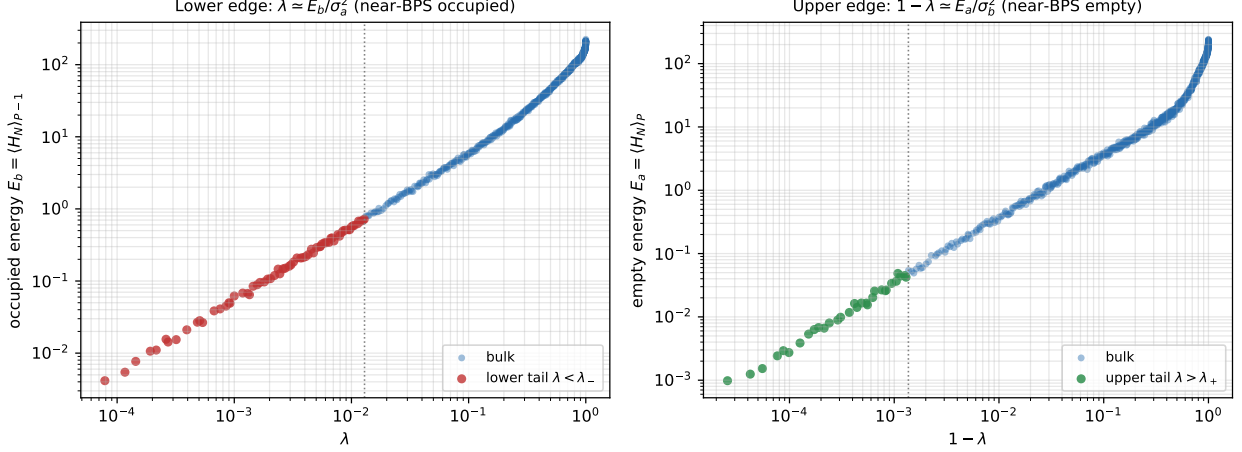


Figure 10: Both tails are near-BPS shadows ($N = 12$). *Left*: the decoder eigenvalue λ versus the energy $E_b = \langle H_N \rangle_{P-1}$ of the state’s N' -occupied component; the lower-tail states (red, $\lambda < \lambda_-$) are the near-BPS ($E_b \rightarrow 0$) modes at degree $P - 1$, and the relation is nearly deterministic and linear, $\lambda \simeq E_b/\sigma_a^2$. *Right*: the mirror— $1 - \lambda$ versus the energy $E_a = \langle H_N \rangle_P$ of the N' -empty component; the upper-tail states (green, $\lambda > \lambda_+$) are the near-BPS ($E_a \rightarrow 0$) modes at degree P , with $1 - \lambda \simeq E_a/\sigma_b^2$. Bulk states (blue) sit at generic energies. The two hard edges carry one near-BPS shadow each.

Equation (41) closes the loop with the energy-side description. The near-BPS density $\rho_H(E)$ near $E = 0$ —a BPS $\delta(E)$ plus a continuum opening above a gap—is exactly the object of the energy-side gravitational (Bessel/near-BPS super-JT) model [9]. The soft tail is thus the smaller theory’s near-BPS spectrum, imaged into overlap variables by the uplift through the resolvent $G = 1/H_N$: *the overlap shadow of the near-extremal throat*, with overlap and energy two probes of the same near-BPS physics related by the linear map (40). This explains every feature of §5. The tail is *macroscopic*—not a dilute eigenvalue instanton—because it is sourced by the near-BPS continuum, an $O(1)$ density, rather than by rare tunneling; it *grows* with N because that continuum grows ($\Gamma \sim e^{S_0}$ increases, the gap shrinks); and its shape has *no fixed power law* because it inherits the ($\sqrt{E - E_0}$ -type) near-gap profile of ρ_H , which drifts with the gap. The wormhole/free-probability sum misses it precisely because free probability treats the projectors as generic (a flat G); the near-BPS enhancement of $1/H_N$ is the structure it discards.

The linear law is a theorem, but its coefficient $c = 1/\langle \sigma_a^2 \rangle$ is not yet closed in form. Its *scale* is fixed by the couplings and the combinatorics—the mean of $D^\dagger D$ on Λ^P is $2 \binom{N-P}{2} \mathbb{1}$, and the tail $|a\rangle$ sits off $\ker(D \wedge)$, so $\langle \sigma_a^2 \rangle = O(P^2)$ —but the exact value is a *constrained* Gaussian average that resists reduction. The BPS conditions fix $|a\rangle$ mostly (94% overlap) along $(Q_N^\dagger)^+ \iota_{\bar{D}} |b\rangle$, yet σ_a^2 is dominated by the small residual set by the second condition: the aligned direction alone gives

$\sigma_a^2 \simeq 39$, whereas the true value is 55, because the residual aligns with the large-eigenvalue directions of $D^\dagger D$. The average is thus fully correlated with D and does not factorize—it matches neither the naive $2\binom{N-P}{2}$ nor the nonzero-subspace mean $(P+1)(P+2)$, and measured $\langle \sigma_a^2 \rangle = 55, 38$ at $N = 12, 13$, so c drifts weakly with N . We leave a closed form open; c is a microscopic constant of scale $O(1/P^2)$, evaluated numerically ($\sigma_a^2 = 55.0 \pm 3.0$ at $N = 12$, giving $c = 0.0182$).

With the law in hand the tail becomes computable entirely on the gravity side: obtain $\rho_H(E)$ from the energy-side (super-JT/Bessel) model, and apply the *derived* linear map $\lambda = cE$ of (38) to get ρ_{tail} through (41). Only the single constant c must still be supplied numerically; the map itself, and the two-shadow structure of the edges, are exact consequences of the BPS constraint.

7 Deviation from Wachter: the finite- N error terms

The soft tail is the statement that at finite N the decoder spectrum departs from the Wachter law. That departure can be made completely explicit. The generic-MANOVA analysis of Kunisky [11] studies precisely the product of two projections ABA , which is our $O_T = P_B \Pi_T P_B$ (dictionary $A = P_B$, $\alpha = a$; $B = \Pi_T$, $\beta = b$; $N = D$), and provides an exact recursion for the finite-size error. Write

$$\text{Tr}(O_T^k) = D m_k^W(a, b) + \Delta_k, \quad (42)$$

with m_k^W the Wachter moment (12) and $D = \binom{N+1}{p}$. Kunisky's error recursion (Lemma 4.6 of [11]) is sourced by three quantities: $\text{Tr } P_B - aD$, $\text{Tr } \Pi_T - bD$, and the *mixed centered traces*

$$T_\ell \equiv \text{Tr} [(P_B - a)(\Pi_T - b)]^\ell, \quad \ell \geq 1. \quad (43)$$

For our decoder the first two vanish *identically*— P_B and Π_T are honest projectors of ranks $d = aD$ and $m = bD$ —so the error is driven by the T_ℓ alone and the recursion is triangular,

$$\tilde{\Delta}_0 = 0, \quad \tilde{\Delta}_k = - \sum_{r=1}^{k-1} \tilde{\Delta}_r \sum_{j=1}^{k-1} (-a)^{k-j} q_{k,j,r} (b^{-1} - 1) + b^{-k} T_k, \quad \Delta_k = b^k \sum_{\ell=0}^k \binom{k}{\ell} \tilde{\Delta}_\ell, \quad (44)$$

with $q_{k,j,r}$ the combinatorial polynomials of [11] (their Definition 4.2). Being triangular with unit diagonal, the system has a unique solution expressing Δ_k as a linear functional of the mixed traces,

$$\Delta_k = \sum_{\ell=1}^k K_{k,\ell}(a, b) T_\ell. \quad (45)$$

We do not evaluate the $q_{k,j,r}$: the coefficients $K_{k,\ell}$ follow in closed form directly from the geometry of the two projectors, as we now show. In the two natural band variables

$$u \equiv a + b - 2ab = \frac{1}{2}(\lambda_- + \lambda_+), \quad v \equiv ab(1-a)(1-b) = \left(\frac{\lambda_+ - \lambda_-}{4} \right)^2, \quad (46)$$

namely

$$K_{k,\ell}(a, b) = \sum_{i \geq 0} \binom{k}{\ell + 2i} \binom{\ell + 2i}{i} u^{k-\ell-2i} v^i. \quad (47)$$

Origin of the closed form. Resolve the two projectors by their cosine–sine decomposition into 2×2 blocks, one per principal angle θ , with $\lambda = \cos^2 \theta$ the decoder eigenvalue on the block. Taking $P_B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\Pi_T = \begin{pmatrix} c^2 & cs \\ cs & s^2 \end{pmatrix}$ ($c = \cos \theta$, $s = \sin \theta$), a direct computation gives, on the block,

$$\text{Tr} [(P_B - a)(\Pi_T - b)] = \lambda - u, \quad \det [(P_B - a)(\Pi_T - b)] = a(1 - a)b(1 - b) = v. \quad (48)$$

Hence the two eigenvalues $\mu_{\pm}$ of $M = (P_B - a)(\Pi_T - b)$ on the block have sum $\lambda - u$ and product v , so

$$T_\ell = \sum_{\text{blocks}} (\mu_+^\ell + \mu_-^\ell) = \sum_{\text{blocks}} D_\ell(\lambda - u, v), \quad D_\ell(s, v) = 2v^{\ell/2} \mathsf{T}_\ell\left(\frac{s}{2\sqrt{v}}\right), \quad (49)$$

where D_ℓ is the Dickson polynomial (the power sum of two roots of given sum and product) and T_ℓ the Chebyshev polynomial. Since $\text{Tr}(O_T^k) = \sum_{\text{blocks}} \lambda^k$, the identity (45) is precisely the block–wise expansion of the monomial in the Dickson/Chebyshev basis, $\lambda^k = \sum_\ell K_{k,\ell} D_\ell(\lambda - u, v) + \text{const}$; the standard monomial–to–Chebyshev inversion applied to $(s + u)^k$ with $s = \lambda - u$ yields (47). Special cases: $K_{k,k} = 1$, $K_{k,k-1} = ku$, $K_{k,k-2} = \binom{k}{2}u^2 + kv$, and at $v \rightarrow 0$ (degenerate band) $K_{k,\ell} = \binom{k}{\ell}u^{k-\ell}$, a shifted binomial transform. As a check that this geometric route agrees with the q –recursion (44), we verified that (47) solves (44) identically for all $k \leq 6$.

Deviation of the moments. In our normalization $m_k = \frac{1}{d} \text{Tr}(O_T^k)$ ($d = aD$), (42)–(45) give the departure of each SYK moment from Wachter in closed form,

$$m_k = m_k^W(a, b) + \frac{1}{a} \sum_{\ell=1}^k K_{k,\ell}(a, b) \tau_\ell, \quad \tau_\ell \equiv \frac{T_\ell}{D} = 2v^{\ell/2} \left\langle \mathsf{T}_\ell\left(\frac{\lambda-u}{2\sqrt{v}}\right) \right\rangle, \quad (50)$$

the τ_ℓ being the Chebyshev moments of the decoder spectrum. On the band $\lambda \in [\lambda_-, \lambda_+]$ the argument lies in $[-1, 1]$ and T_ℓ oscillates; on the soft tail $\lambda \notin [\lambda_-, \lambda_+]$ it exceeds 1 and D_ℓ grows hyperbolically—the tail is the “outside–the–window” part of the Chebyshev moments.

Two normalizations must be kept apart, and they behave oppositely as $N \rightarrow \infty$. The D –normalized traces $\tau_\ell = T_\ell/D$ are themselves $O(a)$: from $\Delta_1 = T_1$, $\Delta_2 = T_2 + 2uT_1$ and the exact identity $\Delta_k/D = a\delta m_k$ (with $\delta m_k \equiv m_k - m_k^W$) one finds $\tau_1 = a\delta m_1$ and $\tau_2 = a(\delta m_2 - 2u\delta m_1)$, so every $\tau_\ell \rightarrow 0$ as the BPS fraction $a(N) \rightarrow 0$ —the mixed traces are diluted away, just as for the Haar/free ensemble. What does *not* vanish is the per–BPS–state deviation it encodes: $\delta m_k = \frac{1}{a} \sum_\ell K_{k,\ell} \tau_\ell$ stays $O(1)$ (numerically $\delta m_2 \simeq 0.03$ – 0.04 and non–decreasing over $N \leq 12$), whereas for a genuinely free pair it would fall off as $O(D^{-2})$. The freeness failure thus survives *per state* even as its D –normalized trace is diluted to zero: this per–state persistence, not any saturation of τ_ℓ , is the quantitative face of the soft tail.

Verification. At $N = 12$ (one realization, $a = 0.425$, $b = 0.538$; measured mixed traces $\tau_1 = -0.004$, $\tau_2 = 0.019$, $\tau_3 = -0.001$, $\tau_4 = 0.001$), the measured moments, the Wachter values, and the closed–form error (50) agree to machine precision:

k	m_k (exact diag.)	m_k^W	error $\frac{1}{a} \sum_\ell K_{k,\ell} \tau_\ell$	sum
1	0.529527	0.538462	−0.008934	0.529527
2	0.430895	0.395519	+0.035376	0.430895
3	0.379510	0.323221	+0.056290	0.379510
4	0.345911	0.277973	+0.067938	0.345911

(agreement to $\sim 10^{-16}$; Fig. 11 runs the same check across all k at $N = 10$). Equations (45)–(50) are the exact, closed–form statement of how finite– N fortuity departs from the free/Wachter law: Wachter plus a band–Chebyshev change of basis (47) acting on the mixed centered traces τ_ℓ .

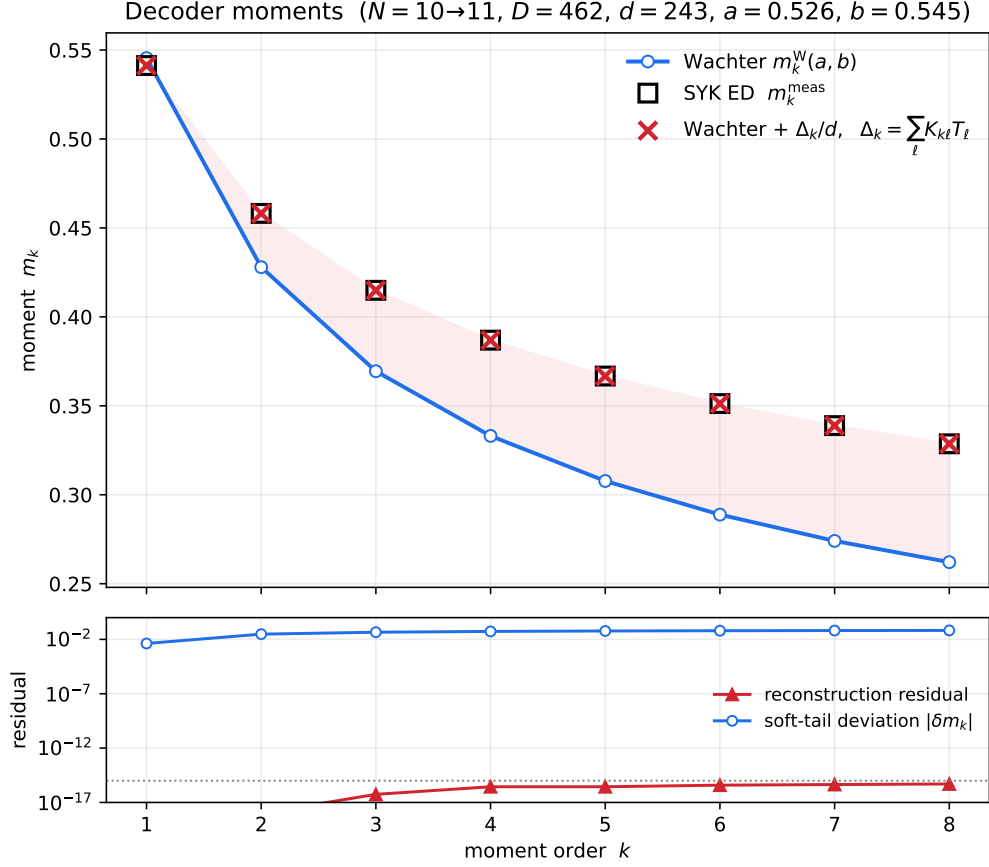


Figure 11: The error formula (50) reproduces the decoder moments exactly. Blue: the Wachter moments $m_k^W(a, b)$, computed with no fitting from the closed-form dimensions ($a = d/D$, $b = m/D$; here $N = 10 \rightarrow 11$, $D = 462$, $d = 243$, $a = 0.526$, $b = 0.545$). Black squares: the SYK decoder moments $m_k^{\text{meas}} = \frac{1}{d} \text{Tr}(O_T^k)$ from exact diagonalization. Red crosses: Wachter *plus* the analytic error, $m_k^W + \Delta_k/d$ with $\Delta_k = \sum_{\ell} K_{k,\ell}(a, b) T_{\ell}$ and the mixed traces $T_{\ell} = \text{Tr}[(P_B - a)(\Pi_T - b)]^{\ell}$ read off the same diagonalization. The crosses land on the squares to machine precision (lower panel: reconstruction residual $\sim 10^{-16}$ at every k), while pure Wachter misses by the soft-tail deviation $\delta m_k = m_k^{\text{meas}} - m_k^W$ (blue, lower panel), which grows from -0.004 at $k = 1$ to $+0.067$ at $k = 8$. The shaded gap is the entire freeness failure, and it is captured completely by the T_{ℓ} .

What the error formula means. Equation (45) separates cleanly into signal and dictionary. The physics sits in the T_{ℓ} . Each is a connected correlation between the BPS subspace and the slot: $(P_B - a)$ is the fluctuation of the BPS projector about its mean density a , $(\Pi_T - b)$ that of the slot about b , so $T_{\ell} = \text{Tr}[(P_B - a)(\Pi_T - b)]^{\ell}$ is a necklace of alternating fluctuations that vanishes identically when the two projectors are free. It measures how strongly the BPS states “know about” the new-mode occupation—the near-BPS alignment that *is* the soft tail. Wachter is the no-signal background; the T_{ℓ} are the signal.

The kernel $K_{k,\ell}$, by contrast, is a change of basis—but a physically natural one. The coordinates adapted to the band $[\lambda_-, \lambda_+]$, centered at u with half-width $2\sqrt{v}$, are the Chebyshev harmonics $T_{\ell}(\frac{\lambda - u}{2\sqrt{v}})$: the normal modes of the band, the analog of Fourier modes on an interval. Equation (47) is the transform between the ordinary powers λ^k (the naive observable) and these harmonics (the

natural one), its two binomials counting the lattice walks of a particle on the band whose on-site energy is the center u and whose hopping amplitude is the width v . Deviations from Wachter are organized by band harmonics; K is the mode-expansion kernel and T_ℓ the occupation of mode ℓ .

The mechanism is clearest in the cosine-sine blocks. The centered product $M = (P_B - a)(\Pi_T - b)$ sends each decoder eigenvalue λ to a pair $\mu_\pm$, the roots of $t^2 - (\lambda - u)t + v$. On the band, $|\lambda - u| < 2\sqrt{v}$ and $\mu_\pm = \sqrt{v}e^{\pm i\phi}$ are conjugate phases on a circle of radius $\sqrt{v}$, with ϕ the position along the band; then $D_\ell = 2v^{\ell/2}\cos(\ell\phi)$ is pure oscillation. On the soft tail, $|\lambda - u| > 2\sqrt{v}$, the pair goes real, $D_\ell \sim 2v^{\ell/2}\cosh(\ell\eta)$ grows, and the eigenvalue has “tunnelled off” the band circle. The tail is exactly the set of states that have tunnelled out, and their hyperbolic growth is why they dominate the high harmonics. In this language the free/Wachter law is the vacuum and the soft tail is a coherent excitation of the band’s normal modes— $K_{k,\ell}$ the mode expansion, T_ℓ the occupations, the tunnelled pairs the excited modes.

Hopping-chain form. The band picture is not a metaphor: it is an exact tight-binding equation, and the walk-counting reading of $K_{k,\ell}$ is literal. The pair $\mu_\pm$ are the roots of $t^2 - (\lambda - u)t + v$, so $\mu_+ + \mu_- = \lambda - u$ and $\mu_+\mu_- = v$; the Dickson power sums $D_\ell = \mu_+^\ell + \mu_-^\ell$ therefore obey Newton’s three-term recurrence

$$D_{\ell+1} = (\lambda - u)D_\ell - vD_{\ell-1}, \quad D_0 = 2, D_1 = \lambda - u. \quad (51)$$

Removing the geometric normalization with $D_\ell = v^{\ell/2}\phi_\ell$ symmetrizes this into a tight-binding Schrödinger equation on a semi-infinite chain of sites $\ell = 0, 1, 2, \dots$,

$$\boxed{\sqrt{v}(\phi_{\ell+1} + \phi_{\ell-1}) + u\phi_\ell = \lambda\phi_\ell}, \quad (52)$$

with hopping amplitude $\sqrt{v}$, on-site energy u , and the decoder eigenvalue λ as the energy. Plane waves $\phi_\ell = e^{i\ell\theta}$ give the dispersion

$$\lambda(\theta) = u + 2\sqrt{v}\cos\theta, \quad (53)$$

so the “band circle” $\mu_\pm = \sqrt{v}e^{\pm i\theta}$ is the Brillouin zone and the propagating band is exactly the Wachter support $[\lambda_-, \lambda_+] = [u - 2\sqrt{v}, u + 2\sqrt{v}]$, edges included. With $H = u\mathcal{K} + \sqrt{v}(S + S^\dagger)$ the shift Hamiltonian on this chain, the k -step amplitude from site 0 to site ℓ reproduces the error kernel exactly,

$$\langle \ell | H^k | 0 \rangle = v^{\ell/2} K_{k,\ell}(u, v), \quad (54)$$

the two binomials of (47) counting the walks with $\ell + i$ forward hops, i backward hops and $k - \ell - 2i$ on-site stays. The soft tail is the chain’s bound-state spectrum: at imaginary momentum $\theta = i\eta$ one has $\lambda = u + 2\sqrt{v}\cosh\eta$ outside the band with an evanescent $\phi_\ell \sim e^{-\eta\ell}$, equivalently a real pole of the diagonal resolvent $G_{00}(\lambda) = [(\lambda - u)^2 - 4v]^{-1/2}$ past its branch points at $\lambda_\pm$ —a bound state pulled off the band by the freeness-breaking overlaps, and a direct echo of the Dyson resolvent of §1. The uniform chain (52) is the free/Wachter reference; the actual decoder measure defines instead a *Jacobi* (Lanczos) chain whose recurrence coefficients approach $(u, \sqrt{v})$ only asymptotically, and the soft tail is precisely the deviation of the near-boundary coefficients from these constants—the operator-growth face of the freeness failure carried by the T_ℓ .

8 The large- N limit: collapse of the Wachter band

The ensemble parameters a, b —and hence the entire Wachter law—have a controlled large- N limit, computable in closed form because they depend only on the BPS and sector dimensions, not on

any diagonalization. The limit reorganizes the soft-tail question in a way the finite- N diagnostics cannot see.

The number of decoder eigenvalues is the enlarged BPS dimension $d = \dim \mathcal{B}_{N'}^p$, at half filling $p = \lfloor (N+1)/2 \rfloor$. From the twisted count $d = \tilde{D}_p - \tilde{D}_{N+1-p-q}$ it collapses to a closed form,

$$d(N) = \begin{cases} 3^{N/2}, & N \text{ even}, \\ 2 \cdot 3^{(N-1)/2}, & N \text{ odd}, \end{cases} \quad S_{\text{BPS}} = \ln d = \frac{N}{2} \ln 3 + O(1), \quad (55)$$

an exact identity (verified for all $N \leq 20$; e.g. $d(12) = 3^6 = 729$, matching the spectrum used throughout). The BPS entropy density is therefore exactly $\frac{1}{2} \ln 3$ —coinciding with the extremal/faulty entropy of (14), now pinned as a closed-form count rather than an asymptotic estimate. Against it sits the sector dimension $D = \binom{N+1}{p} \sim 2^{N+1}/\sqrt{\pi N/2}$, so the BPS *fraction* is exponentially small,

$$a = \frac{d}{D} \sim \frac{1}{2} \sqrt{\frac{\pi N}{2}} \left(\frac{\sqrt{3}}{2} \right)^N \longrightarrow 0, \quad \text{rate } \ln 2 - \frac{1}{2} \ln 3 = 0.14384\dots, \quad b = \frac{m}{D} \rightarrow \frac{1}{2}, \quad (56)$$

the asymptotic matching the exact a to three digits already at $N = 100$. The BPS space is an exponentially small fraction of the charge sector even as it remains exponentially large in absolute size—the ambient-sector and BPS entropies of §9 made quantitative.

Feeding $a \rightarrow 0$, $b \rightarrow \frac{1}{2}$ into the Wachter edges (11) contracts the band onto a single point,

$$\lambda_{\pm} \simeq \frac{1}{2} \pm \sqrt{a}, \quad \lambda_+ - \lambda_- \simeq 2\sqrt{a} \rightarrow 0, \quad (57)$$

(width 0.99 at $N = 12$, 0.06 at $N = 60$, 0.004 at $N = 100$), and because $a < b$ the Wachter atoms vanish identically, $p_0 = p_1 = 0$. The two-projector free-probability law degenerates, $\rho_0 \rightarrow \delta(\lambda - \frac{1}{2})$.

The consequence is not that the tail disappears but that the *bulk* does. As $N \rightarrow \infty$ the generic-overlap spectrum collapses to a measure-zero spike at $\lambda = \frac{1}{2}$, leaving the protected core (atoms at $\lambda = 0, 1$) and the soft tail as the entire nonzero structure—deviations from a delta function rather than from a wide band. The wormhole/free-probability sum, which reproduces the band, thus describes an asymptotically vanishing window, and the physics migrates entirely into what that sum misses. This sharpens rather than settles the tail’s large- N fate: the finite- N “fraction below λ_- ” of §5 does not extrapolate through the collapse, since $\lambda_- \rightarrow \frac{1}{2}$ drags the cut into the middle of the spectrum. The invariant question is the fraction of the $3^{N/2}$ BPS states lying *away* from the collapsing $\frac{1}{2}$ -spike—in the atoms and the genuine tail—and whether it remains $O(1)$. Its moment-space avatar is already visible at finite N : the per-BPS-state deviations $\delta m_k = m_k^{\text{meas}} - m_k^{\text{W}}$ stay $O(1)$ and do not decrease with N (Fig. 12), in contrast to the $O(D^{-2})$ they would show for a genuinely free pair—evidence, though not proof, that the non-spike structure survives the collapse. Settling the fraction itself requires the two spectral inputs still open: the scaling of the linear-law coefficient $c = 1/\langle \sigma_a^2 \rangle$ and the near-BPS density $\rho_H(E)$ of §6, which together fix the tail threshold $E_{\star} = \lambda_-/c$ and the assembled fraction $\Phi(N) = \int_0^{E_{\star}} \rho_H(E) dE$. We record (55)–(57) as the rigorous backbone of the large- N analysis and leave $\Phi(N)$ to the systematic treatment in progress.

Scaling of the mixed traces. Because a, b —and hence the kernel $K_{k,\ell}$ —depend only on (N, p) , the mixed traces factor into an extensive dimension times an intensive shape,

$$\boxed{T_{\ell} = d(N, p) t_{\ell}}, \quad t_{\ell} \equiv \frac{T_{\ell}}{d} = O(1), \quad (58)$$

with $d(N, p) = \dim \mathcal{B}_{N'}^p$, the BPS dimension (55). This is forced by the error identity: dividing $\Delta_k = d \delta m_k = \sum_{\ell} K_{k,\ell} T_{\ell}$ by d gives $\delta m_k = \sum_{\ell} K_{k,\ell} t_{\ell}$, so $t_{\ell} = (K^{-1} \delta m)_{\ell}$ inherits the boundedness

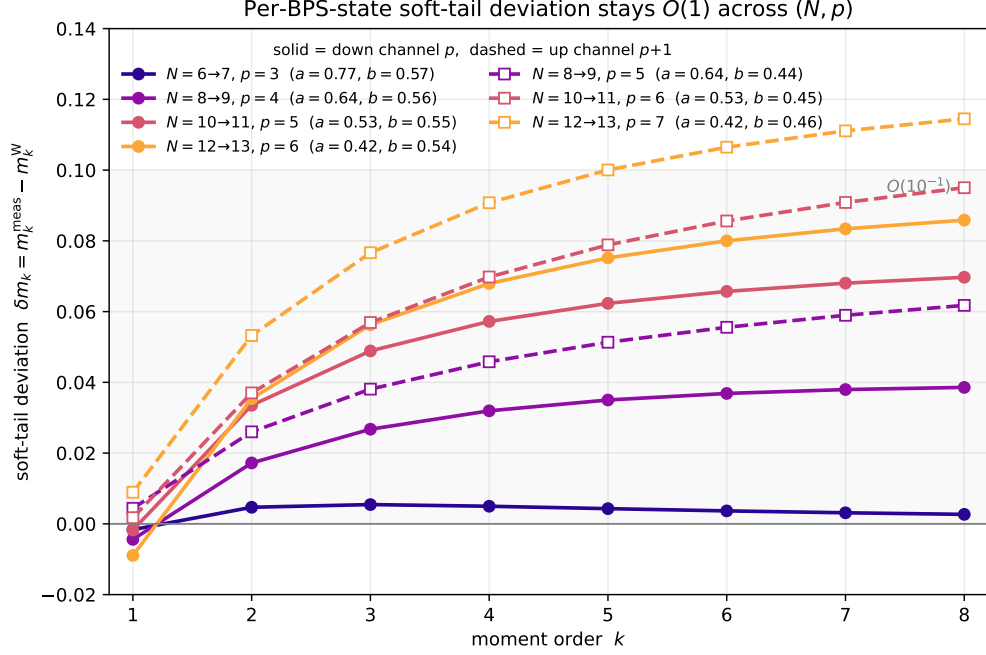


Figure 12: The per-BPS-state soft-tail deviation $\delta m_k = m_k^{\text{meas}} - m_k^{\text{W}}$ versus moment order k , for seven decoder problems of increasing N and both filling channels (solid: down channel $p = \lfloor (N+1)/2 \rfloor$; dashed: up channel $p+1$, same a , mirror b). Each is the gap between exact diagonalization and the closed-form Wachter moments, with no fitting. All curves stay in the $O(10^{-1})$ band and, far from diluting toward zero, *grow mildly and saturate* with N —the signature that the freeness failure is $O(1)$ per BPS state rather than the $O(D^{-2})$ of a free pair. This is the moment-space evidence that structure away from the collapsing $\frac{1}{2}$ -spike persists; it is a finite- N ($N \leq 12$) indication, not a proof of a nonzero limit, which awaits ρ_H .

of the per-state deviations through the triangular, unit-diagonal kernel. Extracted from exact diagonalization, the t_ℓ are small, rapidly ℓ -decaying, identical between the two filling channels, and dominated by $t_2 \simeq \delta m_2$:

(N, p)	a	t_1	t_2	t_3, t_4
(8, 4)	0.64	-0.004	0.022	-0.001, 0.002
(10, 5)	0.53	-0.002	0.035	-0.002, 0.002
(12, 6)	0.42	-0.009	0.044	-0.003, 0.002

($t_{\ell \geq 5} \lesssim 10^{-4}$; the increments of t_2 shrink, 0.015, 0.014, 0.009 over $N = 6, 8, 10, 12$, consistent with saturation to a constant $\simeq 0.05$). Granting $t_\ell = O(1)$, every other quantity scales through the two closed-form factors $d(N, p)$ and $a = d/D$: the traces grow like the BPS dimension, $T_\ell = d t_\ell \sim 3^{N/2}$ (e.g. $T_2 = 0.17, 1.7, 8.6, 32$ tracking $d = 27, 81, 243, 729$); the D -normalized $\tau_\ell = a t_\ell \rightarrow 0$, though only after a pre-asymptotic rise ($\tau_2 = 0.005, 0.014, 0.019, 0.019$, still climbing while t_2 approaches its plateau and a falls); and the per-state deviation $\delta m_k = \sum_\ell K_{k,\ell} t_\ell = O(1)$, dominated by $K_{k,2} t_2$, whence $\Delta_k = d \delta m_k \sim 3^{N/2}$. The lone input not in closed form is thus the $O(1)$ shape t_ℓ —effectively the single number t_2 —whose saturation is evidenced by the shrinking increments but not proven; it is the moment-space image of the near-BPS density ρ_H .

9 Significance and interpretation

An overlap dual, not a geometry dual. The decoder spectrum lives on the compact interval $[0, 1]$ of transmission probabilities, not on a line of energies; correspondingly the bulk object is the Gram matrix of gravitational *overlaps* $\langle \psi_i | \Pi_T | \psi_j \rangle$, and the relevant gravity is the topological ($E = 0$) sector of super-JT with EOW branes. There is no separate dynamical (Weil–Petersson/Schwarzian) content to match *because* we sit at the BPS point where the bulk reduces to exactly this topological brane sum. This is the precise sense in which fragility is dual to a gravitational–overlap computation rather than to spacetime geometry.

Relation to the energy–spectrum models. The chaos of fortuitous BPS states has been characterized on the *energy* side by a universal random–matrix model: a $\Gamma \times \Gamma$ Wishart/Bessel model interpolating to Airy, with the BPS states as exact zeros and a near–BPS continuum above a gap [9, 10, 13, 15]. Our object is the complementary one—the *overlap* Gram matrix of the uplift decoder, whose law is the two–projector Jacobi/Wachter distribution, a two–hard–edge structure (walls at $\lambda = 0$ and $\lambda = 1$) rather than the single hard edge of the Bessel/Wishart energy model. The two share the lesson that fortuity is RMT–universal in the bulk; they differ in what fails *non*–universally. The energy models flag a *dilute* sub–gap eigenvalue leakage; the overlap decoder exhibits the *macroscopic* soft tail of §5, and—because it has two hard edges—a second, mirror tail above the upper edge (§6), one near–BPS shadow per wall. Overlap and energy are thus two faces of fortuitous BPS chaos, universal in the bulk and non–universal in complementary ways at the edge.

Relation to SYK entanglement entropy (an open bridge). The same two–projector Jacobi/MANOVA law appears in the entanglement entropy of the *free* ($q=2$) SYK model [12]: the reduced two–point function $C_A = V^\dagger V$ of an m –mode subsystem, with V a block of the Haar single–particle eigenvector matrix, is exactly $\beta=2$ Jacobi, and its spectrum obeys the Wachter law (11) with $(\kappa, \lambda) = (k/N, m/N)$ in place of our (a, b) . That is the mirror image of our setting: there the eigenvectors are exactly Haar, so Wachter is exact ($T_\ell = 0$) and the interacting entanglement entropy was obtainable only numerically; here the shared couplings make Wachter approximate and the deviation $\delta m_k = \frac{1}{d} \sum_\ell K_{k,\ell} T_\ell$ is the whole content. This points to a bridge worth developing. The slot–side decoder $K_T = \Pi_T P_B \Pi_T$ is the reduced correlation matrix of the maximally–mixed BPS ensemble $\rho = P_B/d$ across the uplift bipartition (old modes $\oplus$ new mode), so the binary–entropy functional

$$S_{\text{ent}} = \sum_i \left[-\lambda_i \ln \lambda_i - (1 - \lambda_i) \ln(1 - \lambda_i) \right] = d \int_0^1 H(\lambda) \rho_{\text{dec}}(\lambda) d\lambda \quad (59)$$

is an entanglement entropy of the BPS states across the added mode—identical in form to the free–SYK result [12] with (a, b) for (κ, λ) . Expanding H in moments splits it as $S_{\text{ent}} = S_{\text{ent}}^W(a, b) + \delta S$ with the soft–tail correction $\delta S = -d \sum_k c_k \delta m_k$ fixed by the *same* error kernel (47); since the tail moves weight toward the walls $\lambda \rightarrow 0, 1$ where $H \rightarrow 0$, one finds $\delta S < 0$. In this language *fragility is an entanglement deficit*: the near–BPS shadow is a set of nearly unentangled directions, and fortuity that is fragile is fortuity whose BPS states are less entangled across the uplift than a generic subspace would be—the overlap analogue of the sub–Page entanglement of interacting SYK, which likewise closes as $q \rightarrow \infty$. Two caveats keep this an outlook rather than a result: reading (59) as a true von Neumann entropy requires the reduced BPS state to be Gaussian (otherwise S_{ent} is its free–fermion proxy), and the appearance of Wachter in SYK entanglement is due to [12]—what

would be new here is the analytic non-free correction δS and its reading as a fragility deficit. We leave both to future work.

Two entropies. The handle expansion is controlled by $1/D = e^{-S_0}$ with $S_0 = \ln D \sim N \ln 2$ the *ambient sector entropy*—not by the BPS entropy $\ln d = \lfloor N/2 \rfloor \ln 3$ (§8). The BPS entropy counts the *branes* (d); the sector entropy weights the *handles* (e^{-2S_0} each). Keeping these two entropies distinct is the whole content of finite- N fortuity: the BPS degeneracy (14) is an $O(1)$ -in- S_0 effect that the naive large- N (planar) gravity misses, while the chaotic bulk is the RMT-universal handle tower.

The protected atoms as the topological piece. The exactly transmitted/lost directions ($\mathcal{B}_N^p \cap \ker C', \mathcal{B}_N^{p-1} \cap \ker C'^\dagger$) are disorder- and handle-independent: the disk diagonal that no wormhole corrects, i.e. the χ (Euler-characteristic) piece of the gravitational sum. But this piece is a *finite- N resonance* rather than a robust index (§1): it is $\sim 5\%$ near $N = 12$ – 13 and vanishes for $N \geq 14$, so at large N even the topology-independent atoms disappear—consistent with the collapse of the band in §8. Gravitationally, Eq. (15) reads

$$\text{fragility} = \underbrace{\text{disk (Wachter)} + \text{wormhole tower}}_{\text{topology-summed, } e^{-S_0} \text{ expansion}} + \underbrace{\text{finite-}N \text{ atom resonance}}_{\text{topology-independent } \chi, 0 \text{ for } N \geq 14}, \quad (60)$$

the robust $\mathbb{Z}_q$ -Witten index $3^{(N-1)/2}$ being a distinct object. Fragility (handle-dependent) and the transient protected atoms (handle-independent) are the two parts of one gravitational overlap sum, the latter fading as $N \rightarrow \infty$.

Chaotic vs. deterministic BPS states. The wormhole/ensemble description applies to the *chaotic* part of the spectrum, and the decoder resolves it. Isolating the properly fortuitous states (projecting out the monotone tower) and pooling over disorder, their level statistics are Gaussian-unitary—adjacent-gap ratio $\langle r \rangle = 0.601$ against the GUE value 0.5996 (Fig. 13)—so the fortuitous states are the chaotic, disorder-random, wormhole-described black-hole microstates. The monotone tower states, by contrast, are built from the coupling-*independent* vector $V^n |\Omega\rangle$ with $V = \sum_i \psi_i^\dagger \chi_i^\dagger$; their decoder eigenvalues are deterministic and do not repel—the non-chaotic, perturbative (“graviton gas”) sector. The discriminant is thus *chaotic vs. deterministic*, not $\lambda = 0$ vs. 1: the chaotic fortuitous states are the ensemble/Wachter bulk gravity captures only statistically, the deterministic tower is the perturbative piece, and the $\lambda = 0, 1$ atoms are the separate topological index.

The soft tail is chaotic too (provisional). The soft tail is non-universal in *density* but not in *correlations*. Restricting the adjacent-gap ratio to the sub-edge states $\lambda < \lambda_-$ alone gives Gaussian-unitary statistics indistinguishable from the bulk: $\langle r \rangle_{\text{tail}} = 0.598 \pm 0.010$ against $\langle r \rangle_{\text{bulk}} = 0.596 \pm 0.005$ and the GUE value 0.603 at $N = 12 \rightarrow 13$ (pooled over 12 realizations, ~ 800 tail ratios), with $\langle r \rangle_{\text{tail}} = 0.584$ at $N = 13 \rightarrow 14$; both lie many σ above GOE (0.536) and nowhere near Poisson (0.386). The tail states are therefore the *same* chaotic spectrum as the bulk—not a separate integrable or near-degenerate sector, which would pull $\langle r \rangle$ toward Poisson—merely pushed below the Wachter edge by the near-BPS enhancement of §6. Physically the soft tail is chaotic black-hole-microstate content imaged into the near-extremal throat, not deterministic “graviton-gas” content. (*Provisional: modest ensembles at $N = 12, 13$; a full N -scan with larger statistics is left to a later version.*)

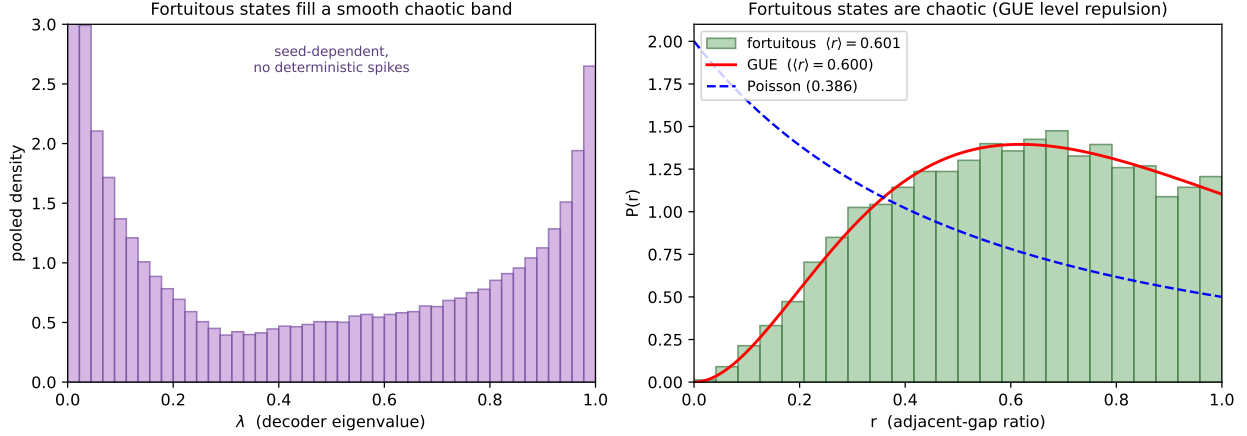


Figure 13: The fortuitous BPS states are chaotic. *Left*: their pooled decoder spectrum is a single smooth (Wachter-like) band with no deterministic spikes—every eigenvalue is disorder-dependent. *Right*: the adjacent-gap ratio distribution matches the GUE surmise (red) and is far from Poisson (blue dashed), $\langle r \rangle = 0.601$. The monotone tower, being coupling-independent, is the deterministic complement removed before this plot.

Status: what is derived, assumed, and open. *Derived*: the microscopic self-energy form (8); the disk law as the planar resummation of brane wormholes (= free probability = Wachter), with Marchenko–Pastur as its dilute limit; the handle coefficients $m_2^{(1)}, m_3^{(1)}, m_4^{(1)}$ from an explicit sum over bulk topologies, resummed into the closed-form genus-one resolvent (31); and (§4) the amplitude dictionary itself from the topological term $e^{S_0 \chi}$ of the super-JT action, which—via the fatgraph duality—upgrades the match to “the action reproduces the handles to all orders” in the BPS sector. *Assumed—and now known to hold only in the bulk*: Haar-typicality of the BPS eigenspace. It reproduces every positive moment, but it *fails macroscopically at the soft edge* (§5): the exact spectrum is not Haar there. So the honest statement is that the gravity/ensemble description is exact for the bulk overlap statistics and the protected index, and breaks down—by an $O(10\%)$ margin—on the near- $\lambda=0$ tail, whose origin we identify as the smaller theory’s near-BPS spectrum imaged by the uplift through the derived identity $\lambda = E/(E + \sigma_a^2)$ (§6). *Open*: the N -scaling of that tail (saturating sector vs. $O(1)$ takeover; §5), a first-principles prediction of the concentration value $\langle \sigma_a^2 \rangle$, and the extension off the BPS point ($E \neq 0$), where the Schwarzian dynamics no longer trivializes.

A How Eq. (31) is obtained, step by step

This appendix expands the one sentence in §3.6 into the actual computation, aimed at a reader who has not used topological recursion. The goal is the genus-one resolvent $W_1(x)$; the only inputs are the disk resolvent $W_0(x)$ and its branch points.

Step 0: the disk resolvent and its curve. The genus-zero resolvent $W_0(x) = \sum_k m_k^{(0)}/x^{k+1}$ is the Stieltjes transform of the Wachter law. Solving the algebraic curve (13) for it gives the explicit two-sheeted function

$$W_0(x) = \frac{1}{x} \left[1 + \frac{x - (a + b) - \sigma(x)}{2a(1 - x)} \right], \quad \sigma(x) = \sqrt{(x - \lambda_-)(x - \lambda_+)}, \quad (61)$$

with $\lambda_{\pm} = \alpha \pm 2\gamma$, $\alpha = a + b - 2ab$, $\gamma = \sqrt{ab(1-a)(1-b)}$, and the branch $\sigma(x) \sim x$ as $x \rightarrow \infty$ (so that $W_0 \sim 1/x$). The two choices of sign of σ are the two “sheets” of the curve.

Step 1: open the cut (Zhukovsky uniformization). Introduce a coordinate z that unfolds the cut into a circle:

$$x(z) = \alpha + \gamma(z + z^{-1}), \quad x'(z) = \gamma(1 - z^{-2}). \quad (62)$$

Then $x'(z) = 0$ exactly at $z = \pm 1$: these two points *are* the branch points $\lambda_{\pm}$. The two sheets become z and $1/z$ (the map $z \mapsto 1/z$ leaves x unchanged), and the square root is now rational: $\sigma(x(z)) = \gamma(z - z^{-1})$.

Step 2: the “y” of the curve. Topological recursion needs how the resolvent differs between the two sheets. From (61), flipping the sign of σ ,

$$y(z) - y(1/z) = W_0^{(+)} - W_0^{(-)} = -\frac{\sigma(x)}{a x(1-x)}. \quad (63)$$

(The poles at $x = 0, 1$ are the Jacobi hard walls; they will matter at the end.)

Step 3: the two universal kernels. Two objects, the same for every one-cut curve:

$$\text{Bergman kernel: } B(z_1, z_2) = \frac{dz_1 dz_2}{(z_1 - z_2)^2}, \quad B(z, 1/z) = -\frac{dz^2}{(z^2 - 1)^2}, \quad (64)$$

$$\text{recursion kernel: } K(z_0, z) = \frac{\frac{1}{2} \left(\frac{1}{z_0 - 1/z} - \frac{1}{z_0 - z} \right) dz_0}{[y(z) - y(1/z)] x'(z) dz}. \quad (65)$$

Step 4: the genus-one one-point function. For $g = 1$, one boundary, the Eynard–Orantin formula is a single residue sum:

$$\omega_{1,1}(z_0) = \sum_{z_* = \pm 1} \text{Res}_{z \rightarrow z_*} K(z_0, z) B(z, 1/z). \quad (66)$$

Step 5: do the residues and simplify. Substituting Steps 2–3, the integrand of (66) has poles of order four at $z = \pm 1$; the residues are elementary (we did them symbolically). Converting the differential to a function via $\omega_{1,1}(z_0) = W_1(x_0) dx_0 = W_1 x'(z_0) dz_0$ and re-expressing in x_0 , two collapses occur: the degree-six denominator factorizes because $(\alpha - x)^2 - 4\gamma^2 = \sigma(x)^2$, giving $-\sigma^6$, and the numerator reduces to a multiple of $x(x-1)$. The result is $\propto x(x-1)/\sigma^5$.

Step 6: fix the normalization. Topological recursion is naturally normalized to the resolvent of the full $D \times D$ matrix $(\frac{1}{D} \text{Tr})$, whereas our moments use the compression $(\frac{1}{d} \text{Tr})$ on the d -dimensional BPS block). The two differ by one overall constant, which is $-1/a^2$; we pinned it exactly by matching at six independent (a, b) values (equivalently, from the exact identity $\mathbb{E}[\text{Tr } O_T^{-1}] = d(D-d)/(m-d)$). This yields Eq. (31),

$$W_1(x) = -\frac{\gamma^2 x(x-1)}{a \sigma(x)^5} = -\frac{b(1-a)(1-b) x(x-1)}{[(x - \lambda_-)(x - \lambda_+)]^{5/2}}.$$

Step 7: check it. Expand $W_1(x) = \sum_{k \geq 2} m_k^{(1)}/x^{k+1}$ at large x : the coefficients reproduce $m_2^{(1)}, m_3^{(1)}, m_4^{(1)}$ of (19)–(21) identically and predict $m_5^{(1)}$. The pole at $x = \lambda_{\pm}$ is of order $6g-4+2n = 4$ in $\omega_{1,1}$ with no residue, the universal signature of a genuine genus-one one-point function; and the numerator’s zeros at $x = 0, 1$ (from the walls in (63)) are why the handle never touches the protected atoms.

B The path integrals, written out

Here are the expressions computing the moments $\overline{m_k}$, from the boundary object we actually evaluate to the formal bulk path integral it is dual to. We flag what is exact and what is formal.

B.1 The exact boundary integral. Averaging the BPS subspace over the couplings makes it Haar-random in the D -dimensional sector, so

$$\overline{m_k} = \frac{1}{d} \overline{\text{Tr}(P_B \Pi_T)^k} = \frac{1}{d} \int_{U(D)} dU \text{Tr}[(U P_0 U^\dagger) \Pi_0]^k, \quad (67)$$

with dU the Haar measure and $P_0 = \text{diag}(\mathbb{1}_d, 0)$, $\Pi_0 = \text{diag}(\mathbb{1}_m, 0)$ fixed projectors of ranks d (BPS) and m (slot). Weingarten calculus evaluates (67) to the closed form (16). *This is exact* (given typicality) and is what the bulk of the paper computes.

B.2 The formal bulk path integral. The same number is the gravitational path integral with the boundary fixed to the k -necklace $\mathcal{N}_k$ —one boundary circle carrying $2k$ alternating EOW-brane insertions (k of species ψ , k of species t , the resolutions of P_B and Π_T)—summed over all bulk fillings:

$$\overline{m_k} = \frac{1}{d} \sum_{\Sigma: \partial\Sigma = \mathcal{N}_k} \int \frac{\mathcal{D}g \mathcal{D}\Phi \mathcal{D}[\text{fermions}]}{\text{Vol}(\text{diff})} e^{-I[\Sigma]}, \quad (68)$$

$$I = -S_0 \chi(\Sigma) - \frac{1}{2} \int_{\Sigma} \sqrt{g} \Phi (R+2) - \int_{\partial\Sigma_{\text{asy}}} \sqrt{h} \Phi_b (K-1) + \mu \int_{\text{brane}} \sqrt{h} + (\text{SUSY}).$$

B.3 Topological reduction. Integrating out the dilaton Φ forces $R = -2$ and reduces each asymptotic boundary to the $\mathcal{N}=2$ super-Schwarzian, whose density of states is $\rho(E) = e^{S_0} \delta(E) + \rho_{\text{cont}}(E)$. The decoder lives at $E = 0$, so every boundary is projected onto the $\delta(E)$ of weight e^{S_0} ; the Schwarzian dressing collapses and the amplitude of a surface becomes its topological value times brane index sums:

$$\boxed{\overline{m_k} = \frac{1}{d} \sum_{\Sigma: \partial\Sigma = \mathcal{N}_k} e^{S_0 \chi(\Sigma)} d^{L_\psi(\Sigma)} m^{L_t(\Sigma)}} \quad (69)$$

where L_ψ, L_t count closed ψ - and t -brane loops and $e^{S_0} = D$. This boxed line is the path integral “we are doing” in the topological sector.

B.4 Why B.3 = B.1. Each filling Σ is a ribbon graph; the ’t Hooft duality gives $D^{\chi(\Sigma)} = e^{S_0 \chi}$, so summing (69) over Σ reproduces (16)=(67) term by term. Organized by genus, $\overline{m_k} = \sum_g e^{-2gS_0} m_k^{(g)}$, and the resolvent $W(x) = \sum_g e^{-2gS_0} W_g(x)$ with $W_0 = \text{Wachter}$ and W_1 from Appendix A.

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